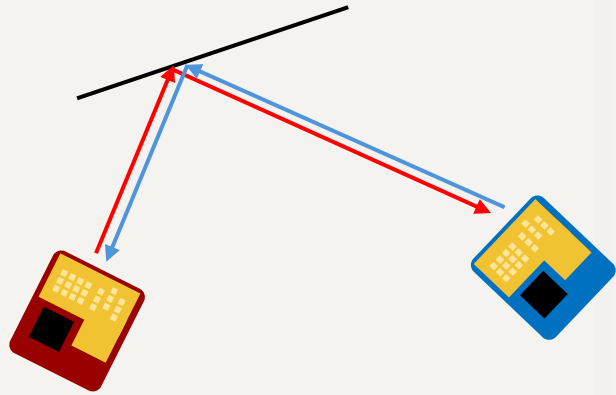
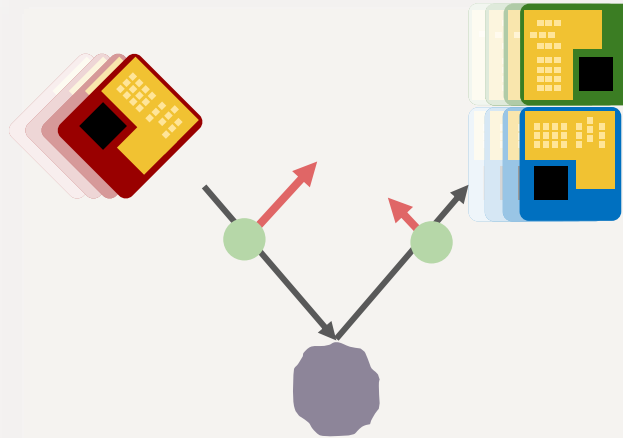


Towards Practical mmWave Sensing: A Distributed and Coherent Radar Approach

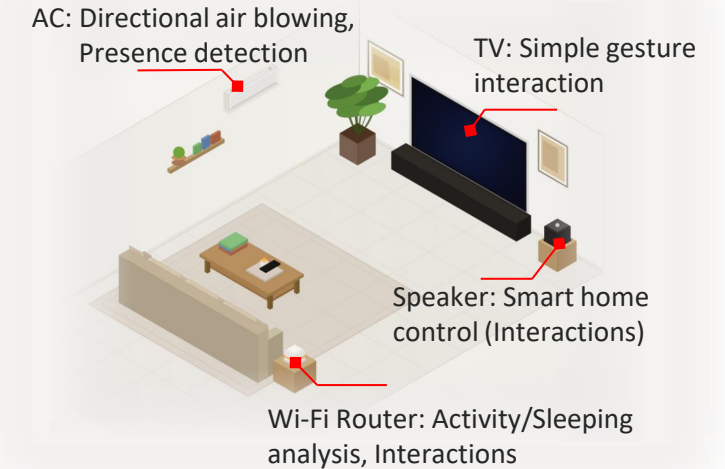
Xinghua Sun (University of Washington)



**I. Improved Information with
Distributed Radar**
MulDar (MobiSys '26)



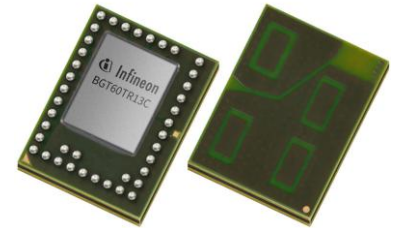
**II. Extended Ability with
Distributed Radar**
POLySight (SenSys '26)



**III. Practicality with
Distributed Radar**
wirelessMulDar (ongoing)



History of mmWave Radars



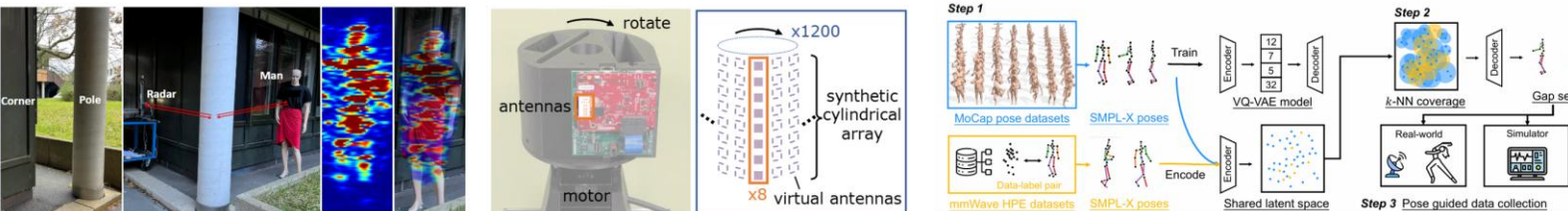
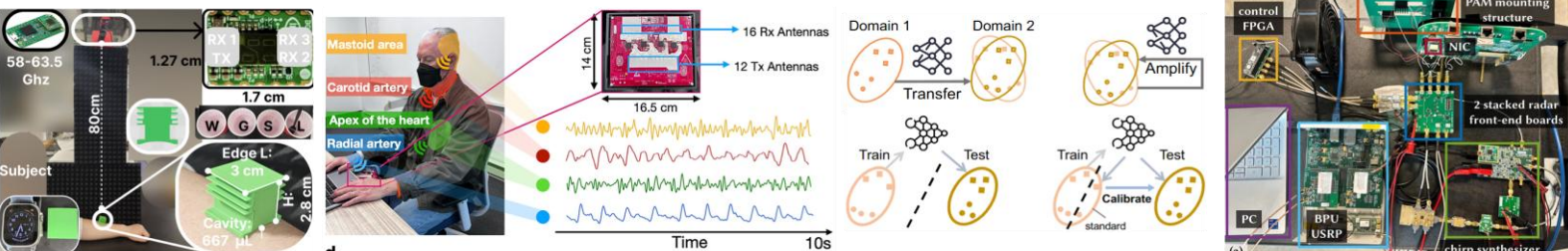
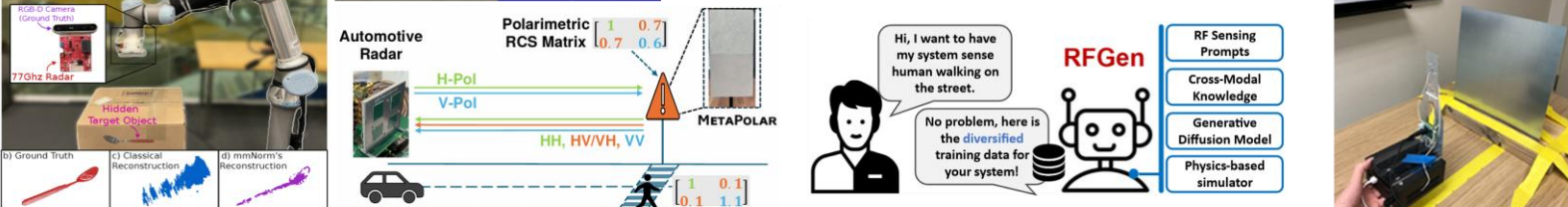
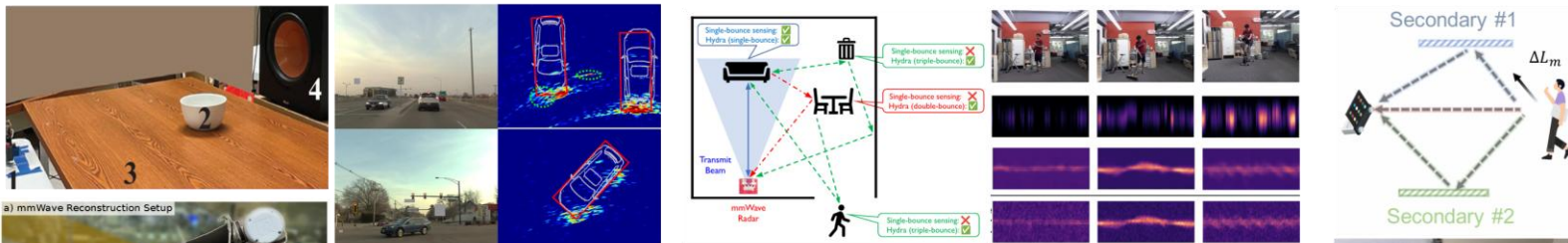
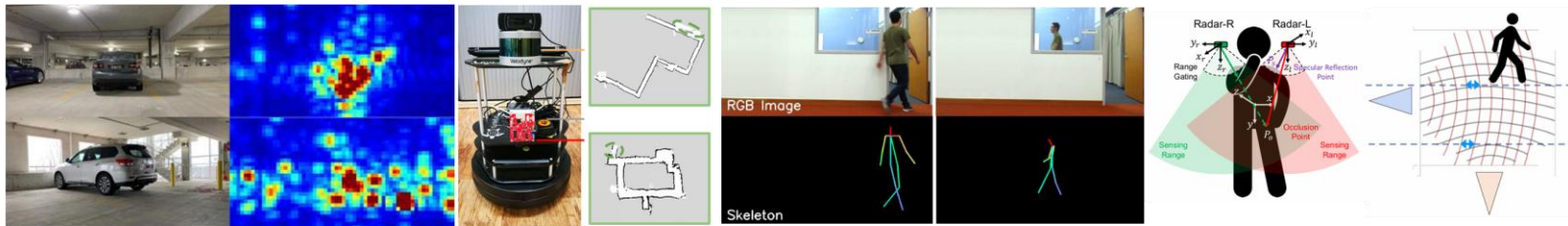
[CloudSat - NASA Science](#)

[Rosemount™ 3408 Level Transmitter - Non-Contacting Radar - Measurement Instrumentation | Emerson](#)

[Soli Radar-Based Perception and Interaction in Pixel 4](#)

[AWR2243BOOST Evaluation board | TI.com](#)

[BGT60TR13C - 60 GHz radar sensors for IoT | Infineon Technologies](#)



3D Shape Reconstruction from Autonomous Driving Radars, BMVC '25

Radatron: Accurate Detection Using Multi-Resolution Cascaded MIMO Radar, ECCV '22

Through Fog High Resolution Imaging Using Millimeter Wave Radar, CVPR '20

See through smoke: robust indoor mapping with low-cost mmWave radar, MobiSys '20

Through-Wall Human Pose Estimation Using Radio Signals, CVPR '18

Vid2Doppler: Synthesizing Doppler Radar Data from Videos for Training Privacy-Preserving Activity Recognition, CHI '21

SeRadar: Embracing Secondary Reflections for Human Sensing with mmWave Radar, MobiCom '25

Argus: Multi-View Egocentric Human Mesh Reconstruction Based on Stripped-Down Wearable mmWave Add-on, SenSys '25

MVDoppler-Pose: Multi-Modal Multi-View mmWave Sensing for Long-Distance Self-Occluded Human Walking Pose Estimation, CVPR '25

Hydra: Exploiting Multi-Bounce Scattering for Beyond-Field-of-View mmWave Radar, MobiCom '24

Contactless material identification with millimeter wave vibrometry, MobiCom '23

mSense: Towards Mobile Material Sensing with a Single Millimeter Wave Radio. Ubicomp '20

Meta-Sticker: Sub-Terahertz Metamaterial Stickers for Non-Invasive Mobile Food Sensing, Sensys '23

MetaBioLiq: A Wearable Passive Metasurface Aided mmWave Sensing Platform for Bio Fluids, MobiCom '24

RF Genesis: Zero-Shot Generalization of mmWave Sensing through Simulation-Based Data Synthesis and Generative Diffusion Models, SenSys '23

MetaPolar: A Low-Cost, Passive, mmWave Radar-Readable Metasurface Road Sign, SenSys '26

Enhancing mmWave Radar Sensing Using a Phased-MIMO Architecture, MobiSys '24

Measuring multi-site pulse transit time with an AI-enabled mmWave radar, Nature Communications '26

Non-Line-of-Sight 3D Object Reconstruction via mmWave Surface Normal Estimation, MobiSys '25

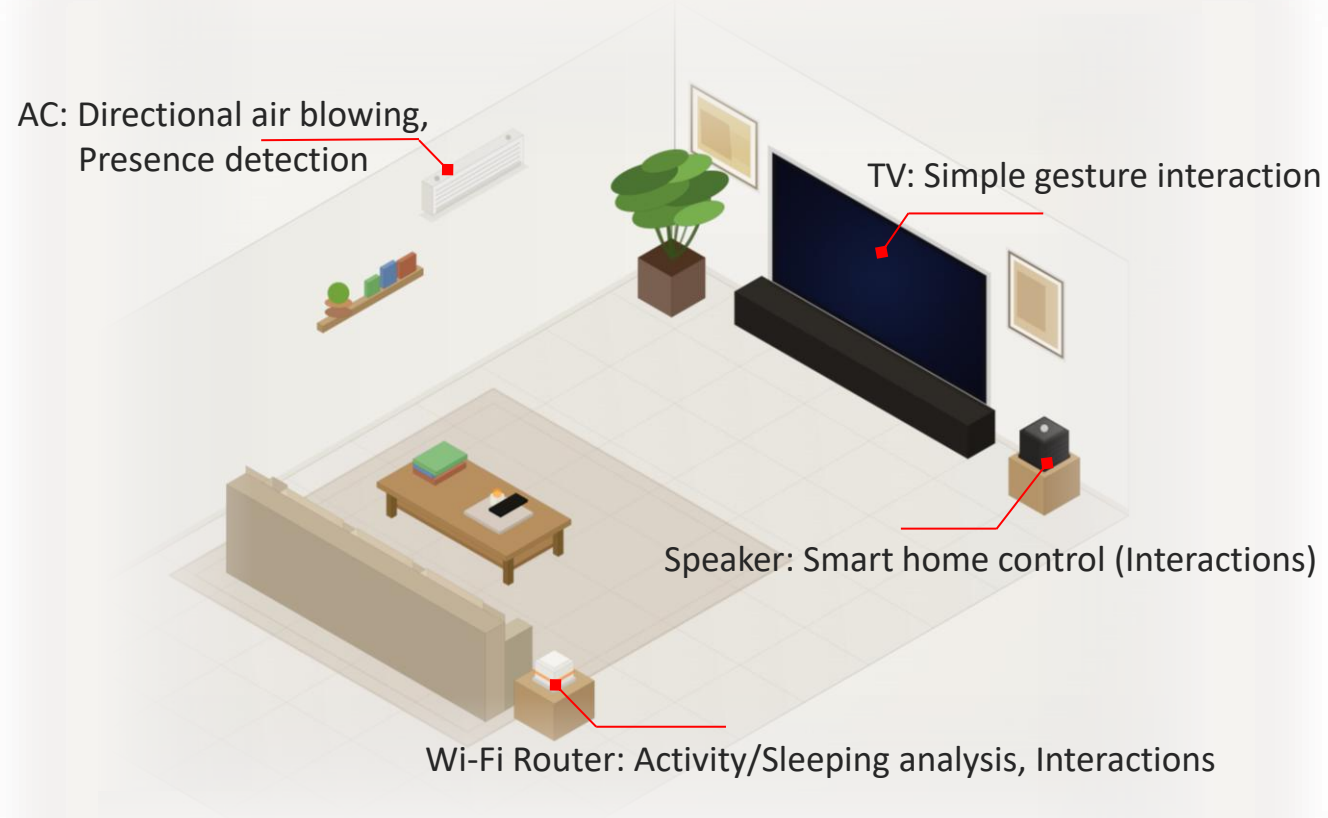
Around the Corner mmWave Imaging in Practical Environments, MobiCom '24

Enabling Visual Recognition at Radio Frequency, MobiCom '24

Data Can Speak for Itself: Quality-guided Utilization of Wireless Synthetic Data, MobiSys '25

WiCompass: Oracle-driven Data Scaling for mmWave Human Pose Estimation, MobiCom '26

Research to Industry Gap

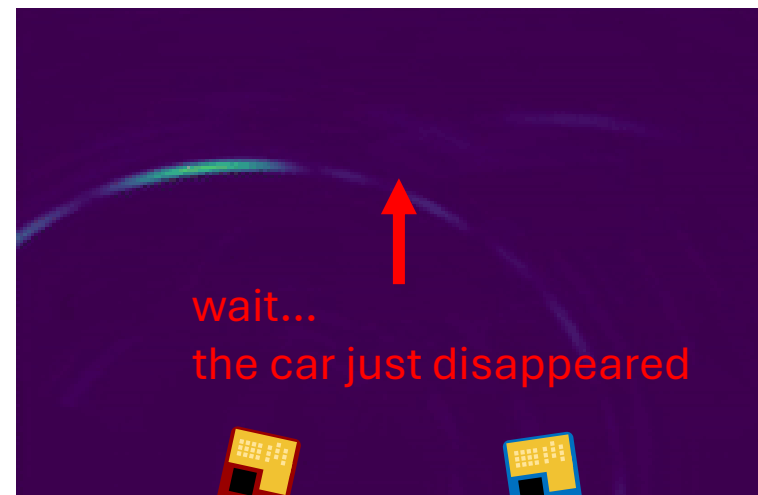


A Simple Experiment



Radar 1

Radar 2

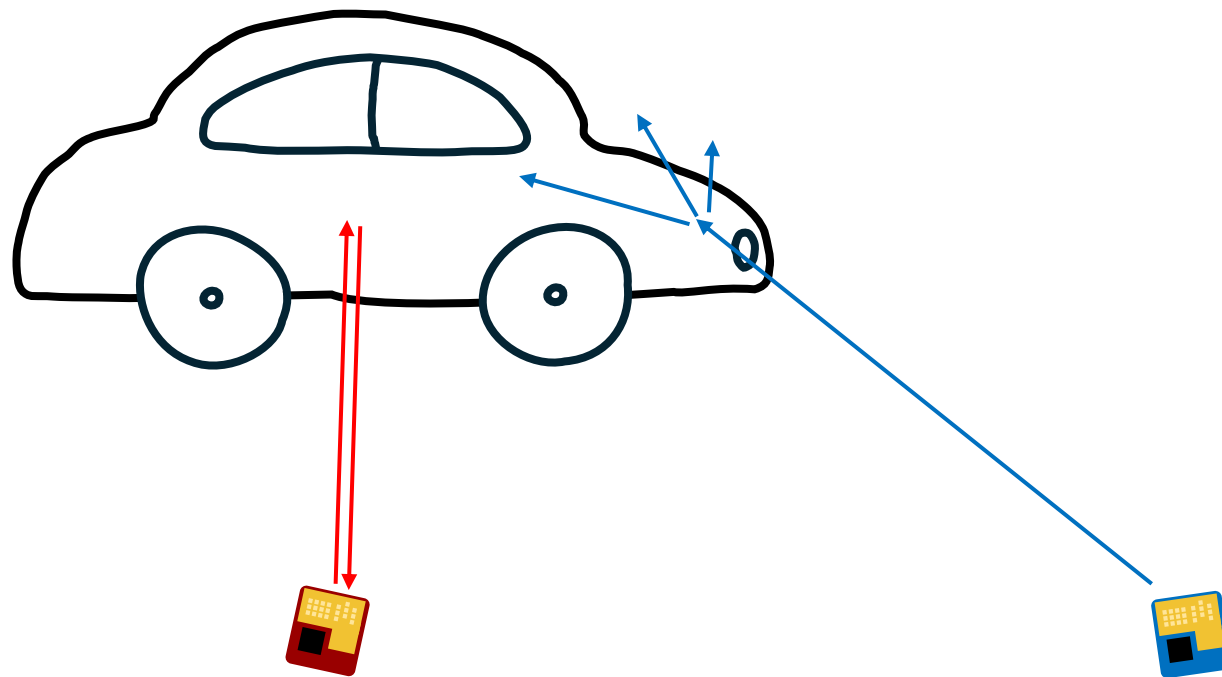


Combined Beamforming
Image of Radar 1 & 2

A Simple Experiment



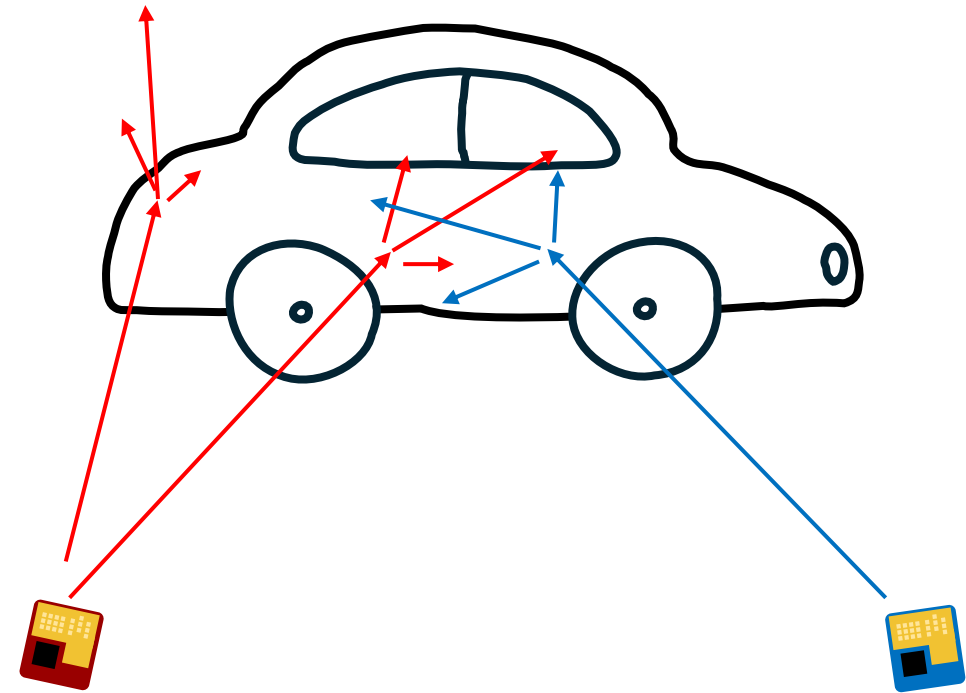
Combined Beamforming
Image of Radar 1 & 2



A Simple Experiment

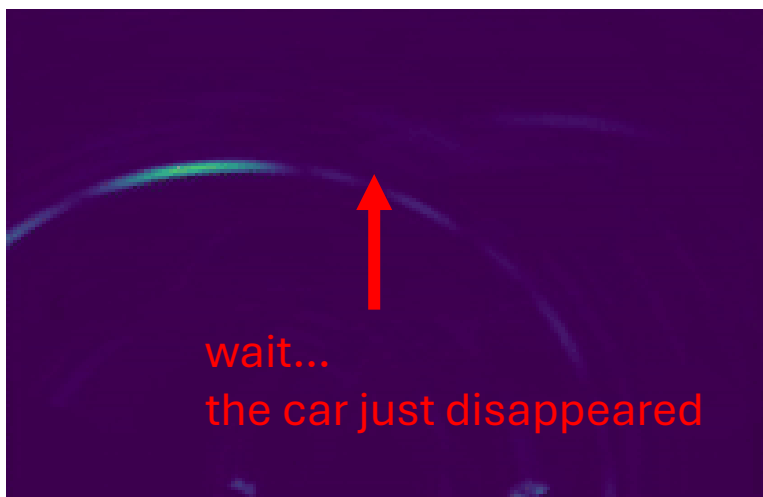


Combined Beamforming
Image of Radar 1 & 2

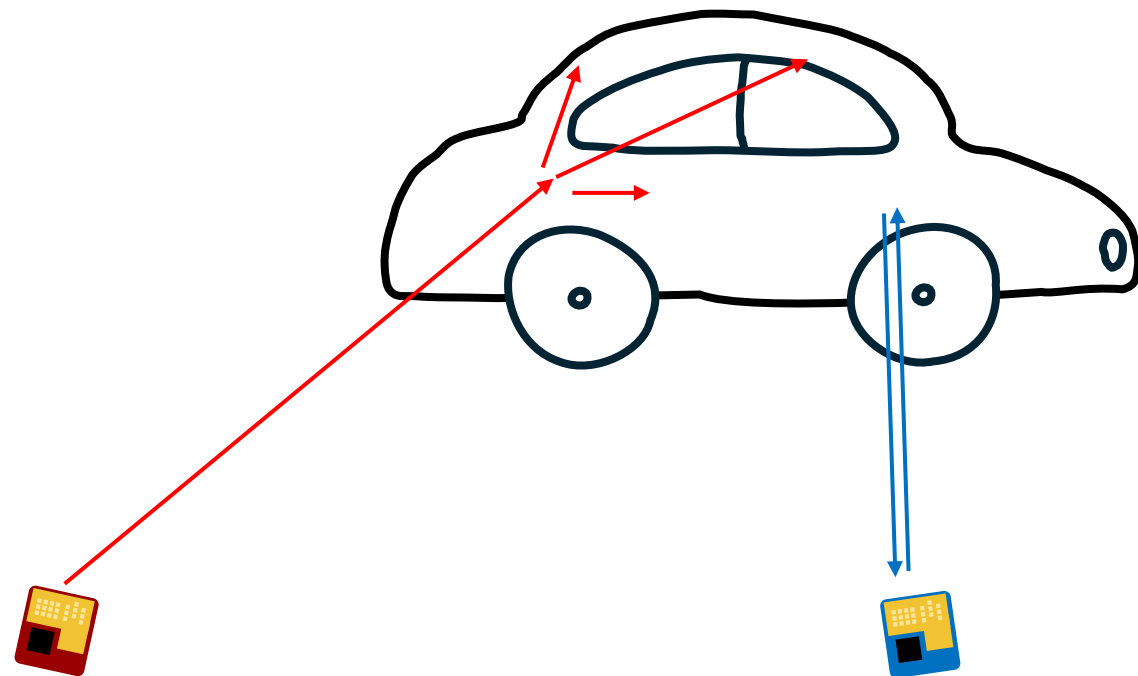


Specular Reflections:
the car is invisible to both radar!

A Simple Experiment



Combined Beamforming
Image of Radar 1 & 2

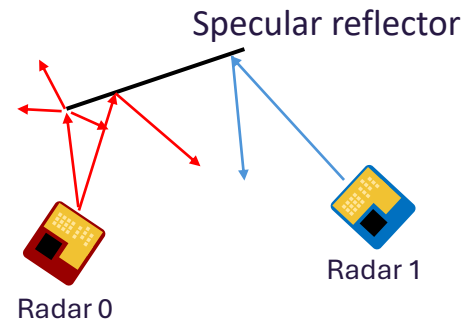


Fundamental Limitations of mmWave Radars

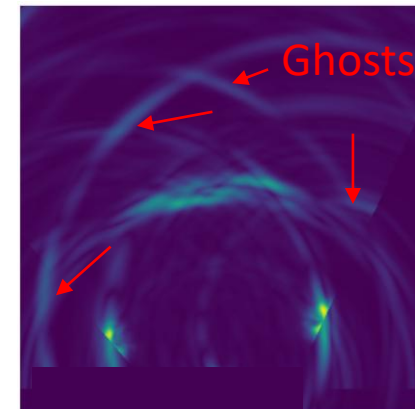
Sparse and noisy Information



Vague and incomplete shape

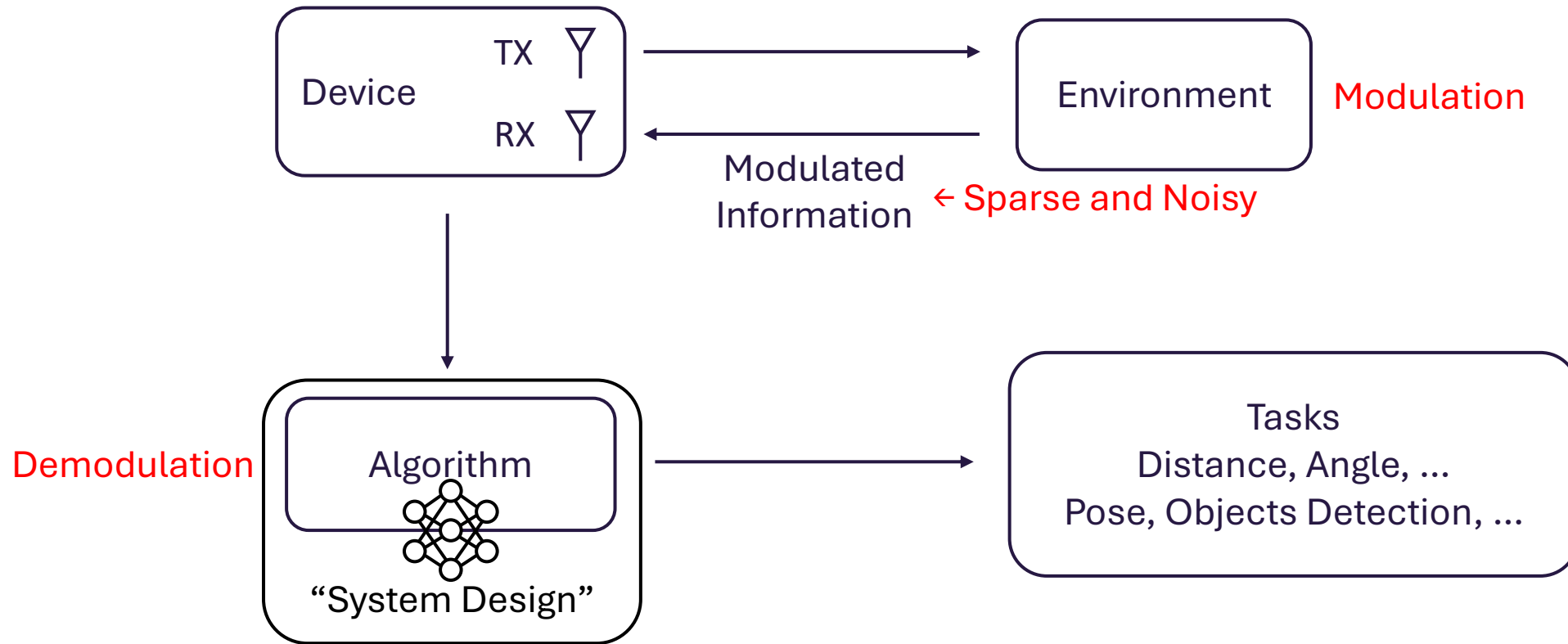


Invisible items

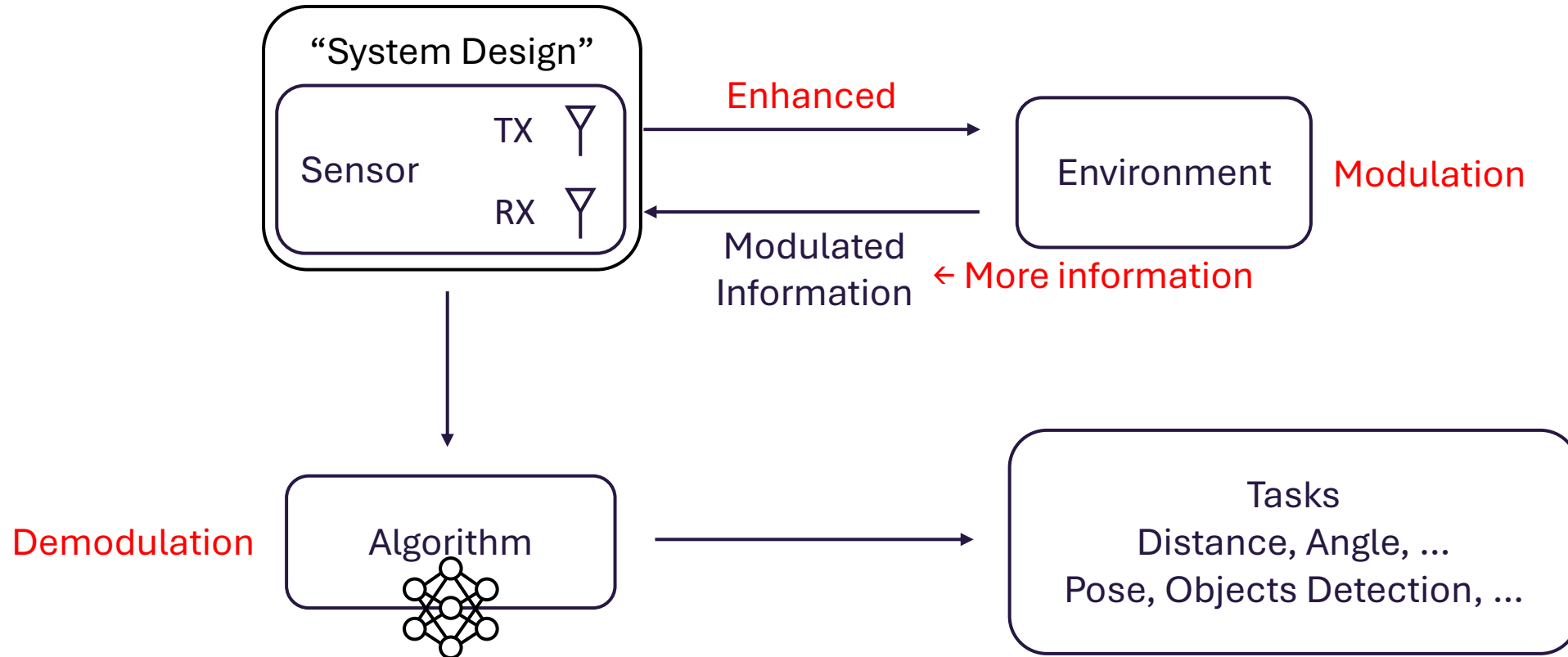


Ghost responses

Sensing is Non-cooperative Communication

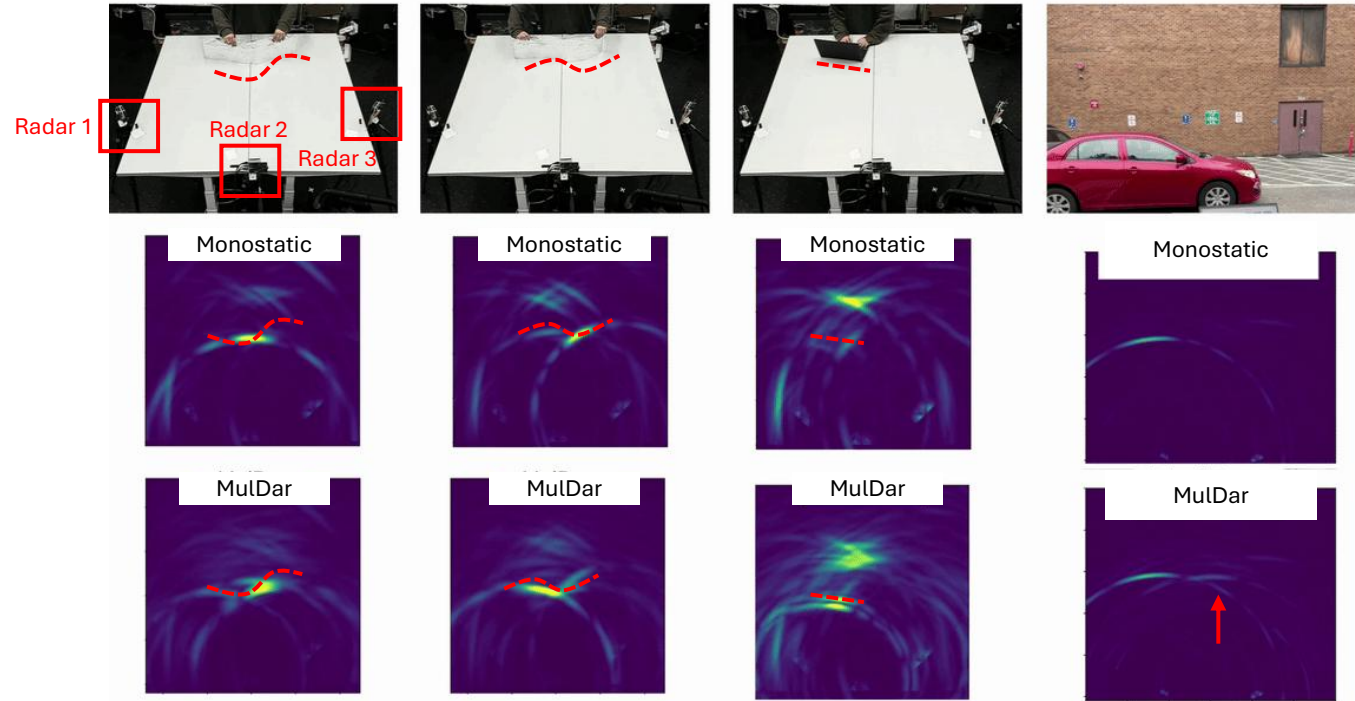


Sensing is Non-cooperative Communication

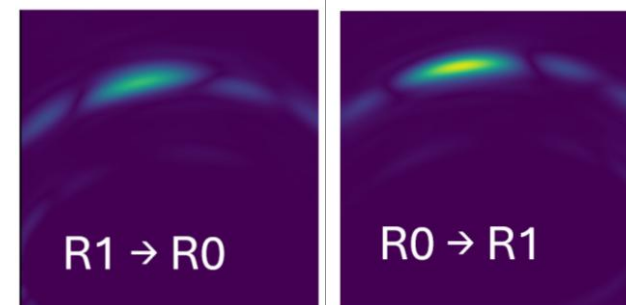
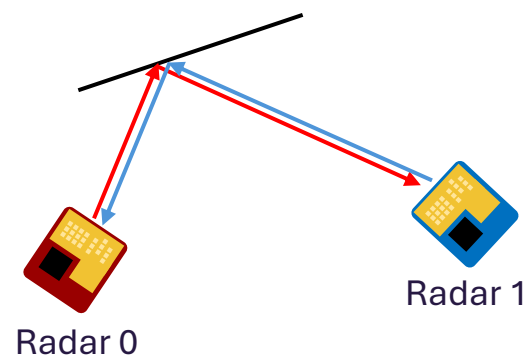
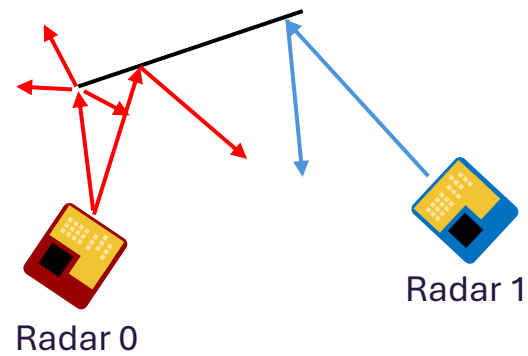


I. Improved Information with Distributed Radar

MulDar: Unleashing the Potential of Distributed COTS mmWave Radar by Exploiting Cross-Device Channels
(MobiSys 2026)

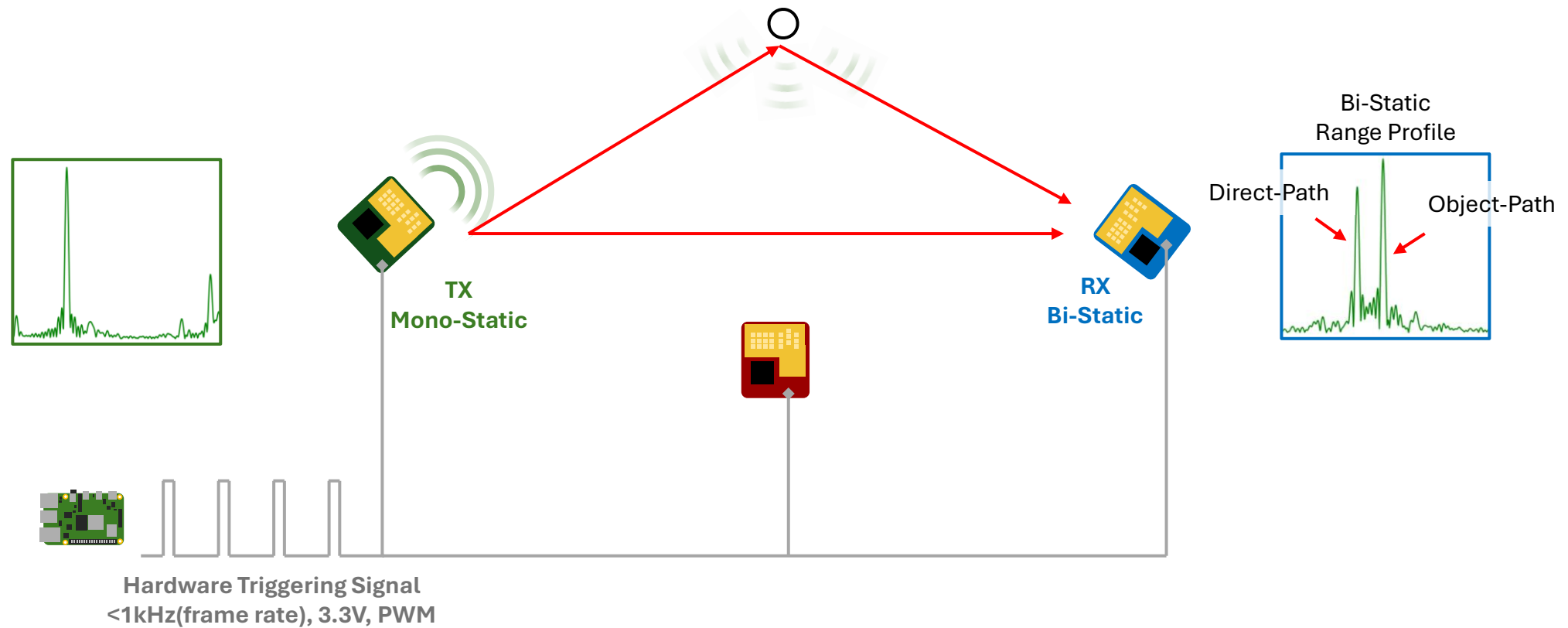


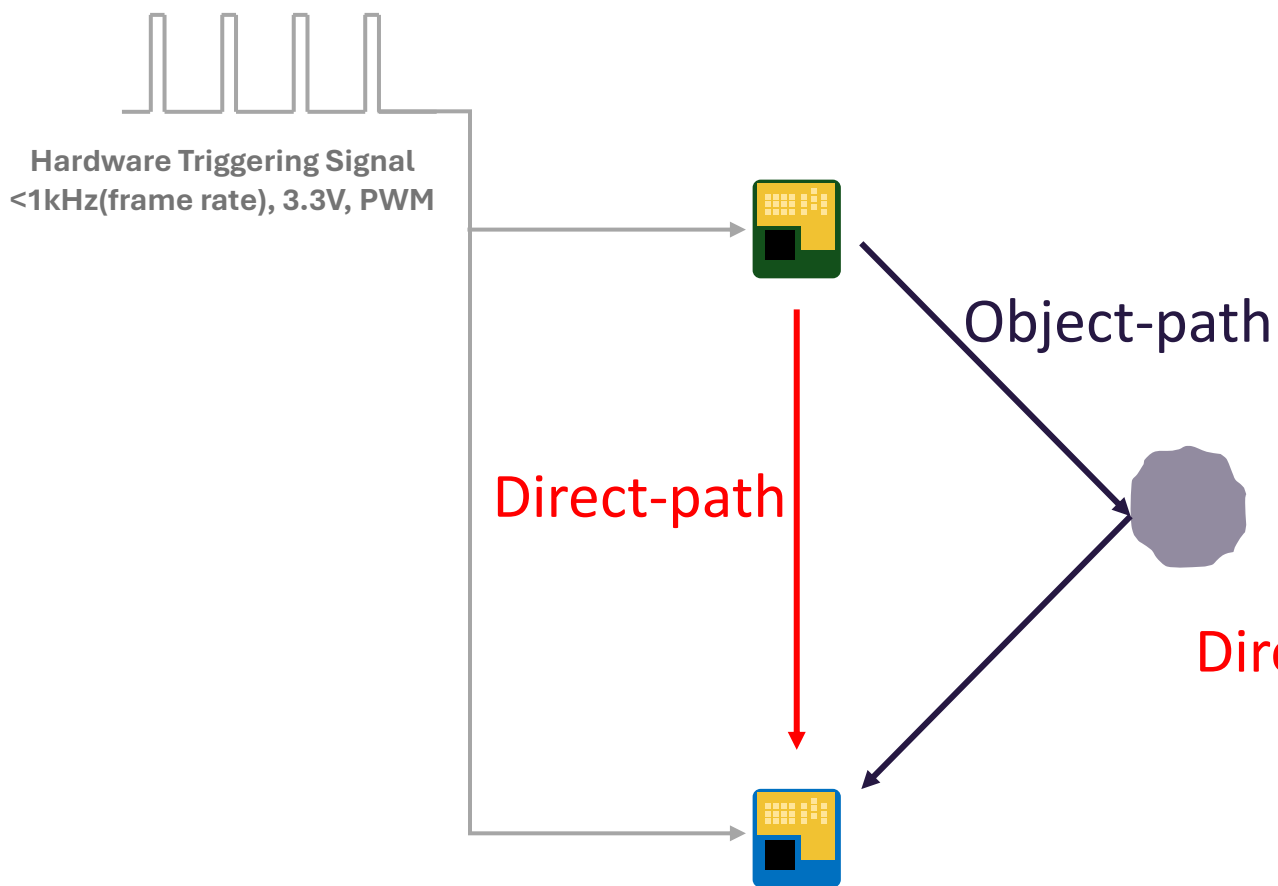
MulDar: A Distributed and Coherent Radar System



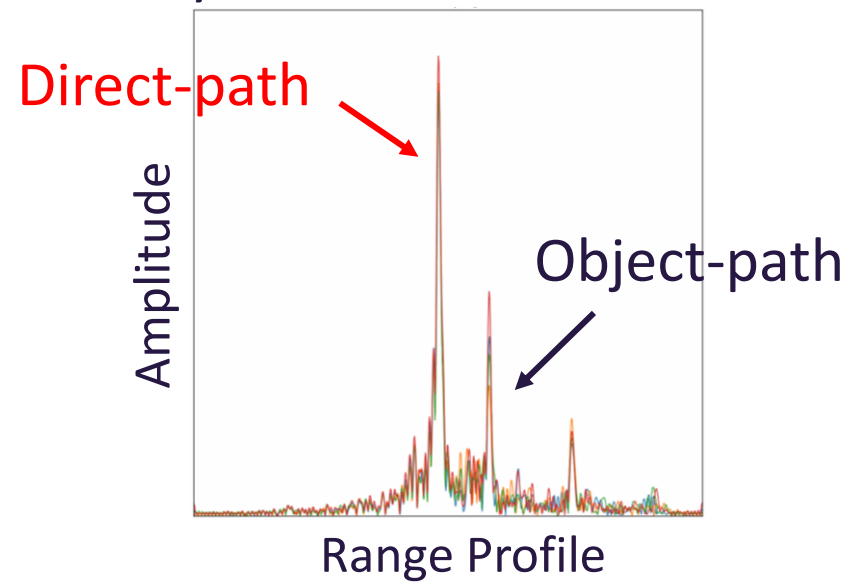
cross-device channel provides more information

Synchronizing 77GHz FMCW Radars

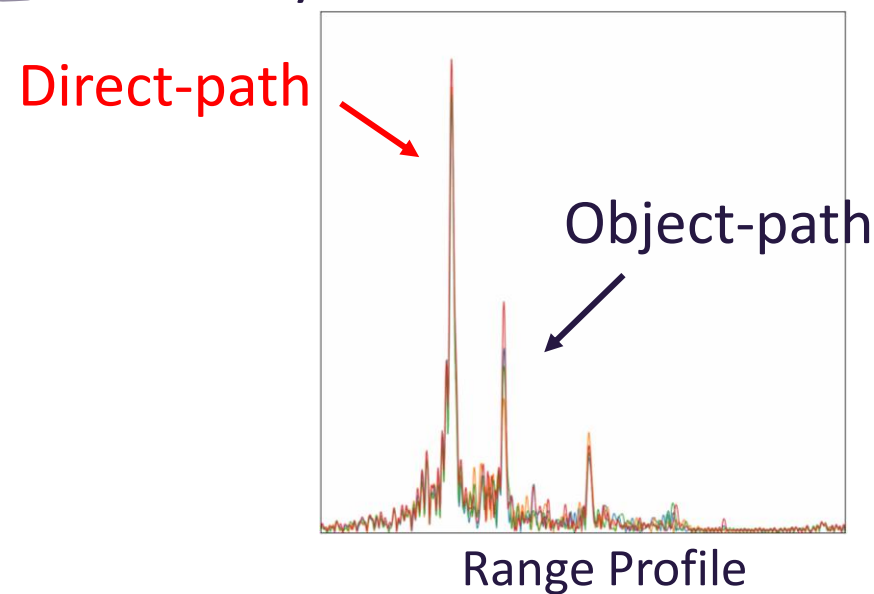




Unsynced Bi-static Channel



Synced Bi-static Channel



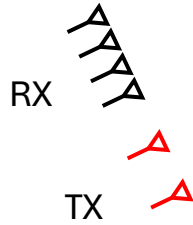
Multi-static Radar Array



Multi-static Radar Array

Distributed radars are combined as one big multi-static radar

No concept of radars, but **antennas**

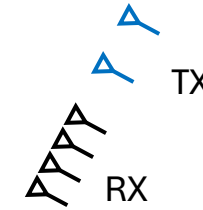


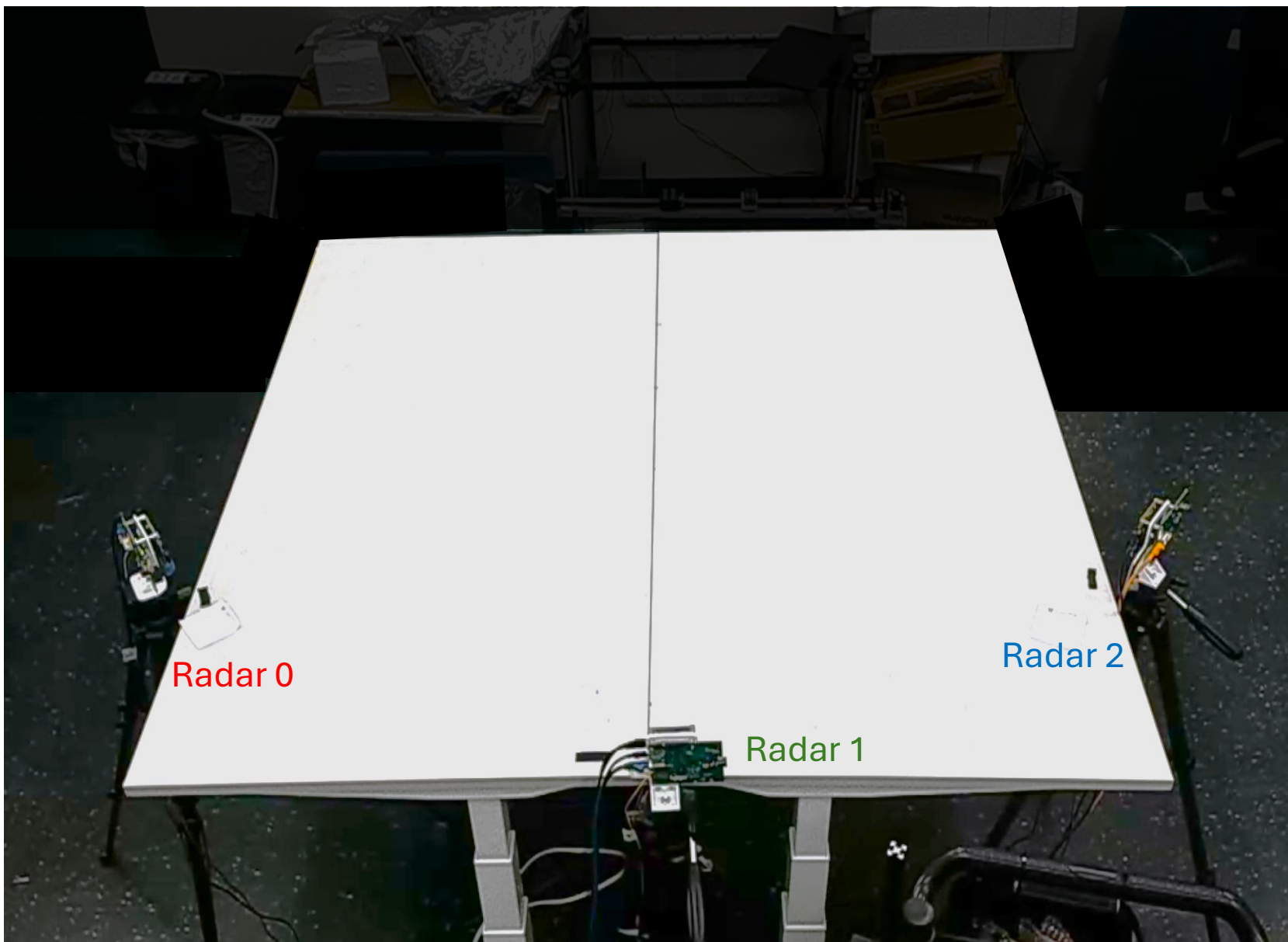
Increase number of channels from n to n^2

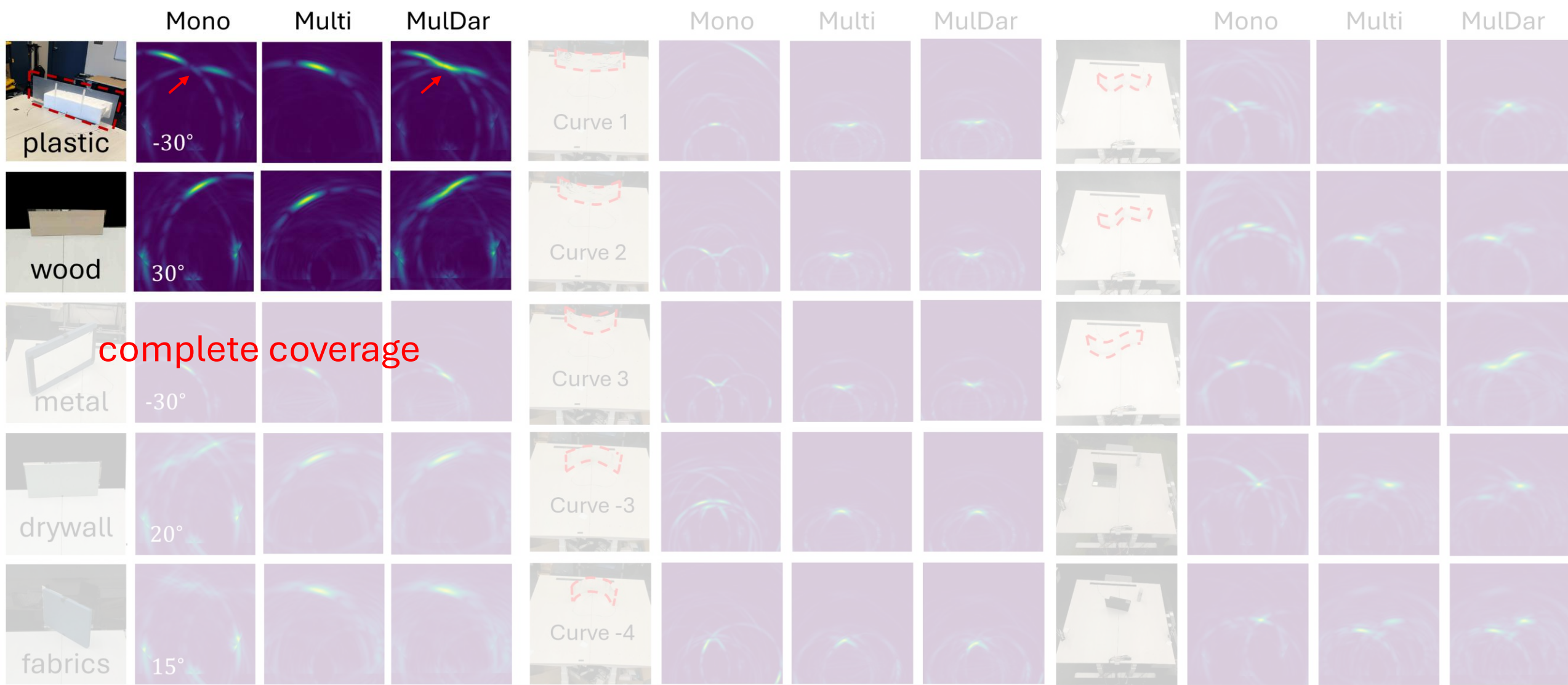
→ much more additional information

Additional perspectives

→ deal with specular reflection



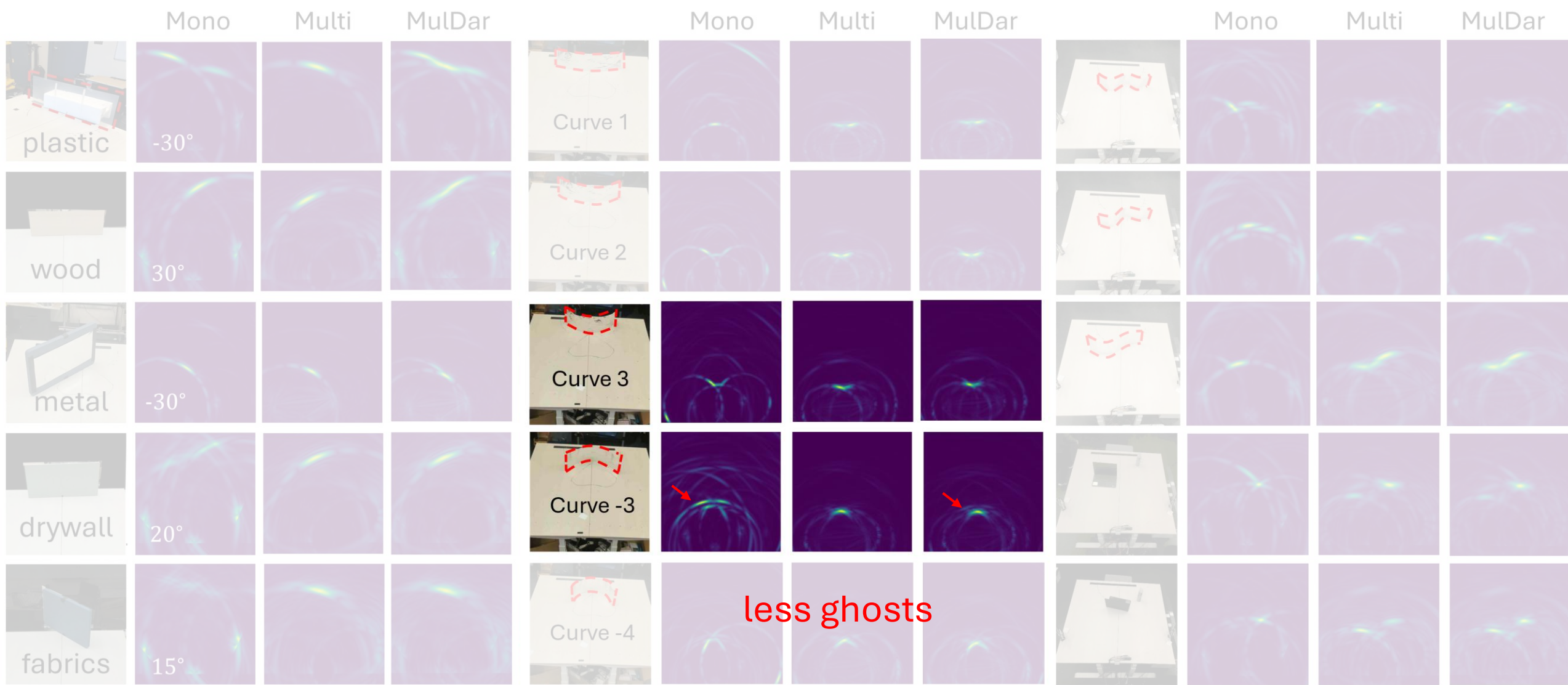




(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes

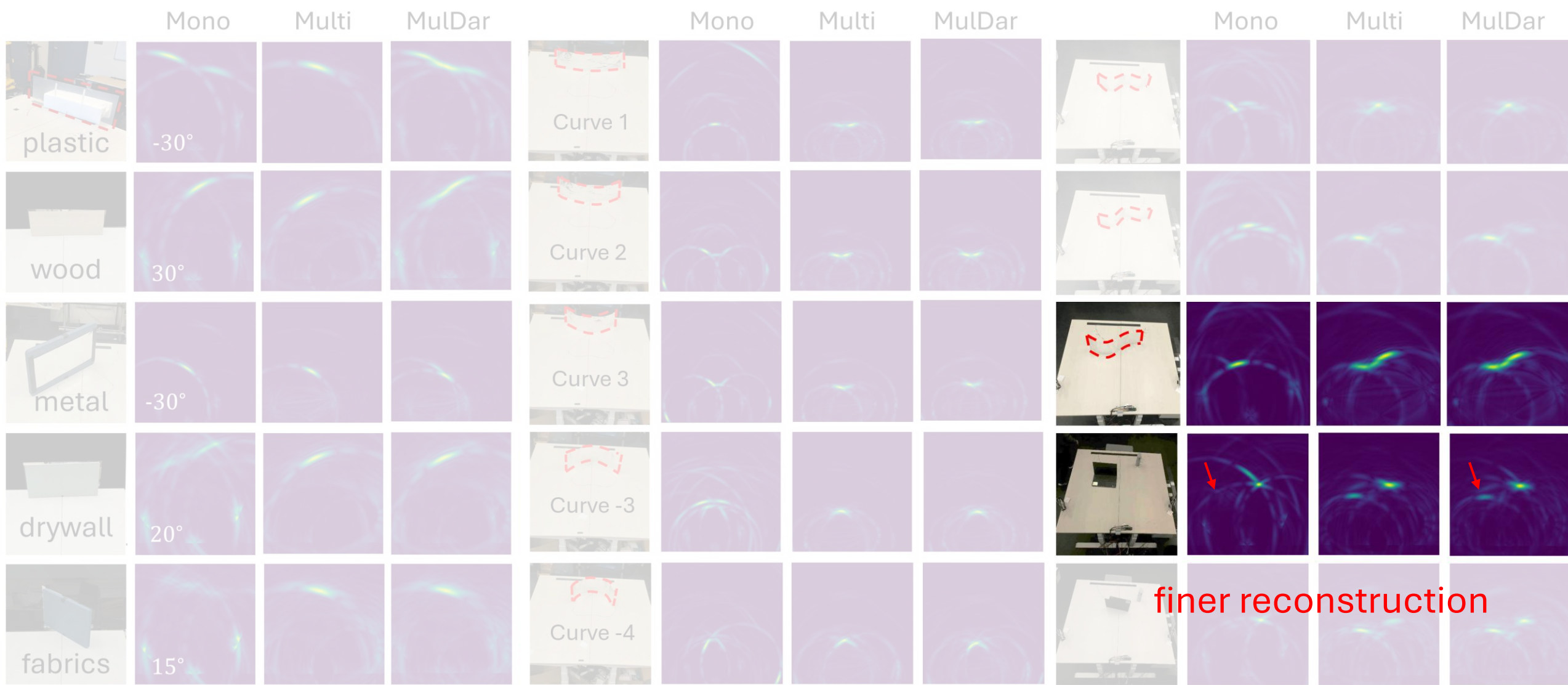


(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes

less ghosts

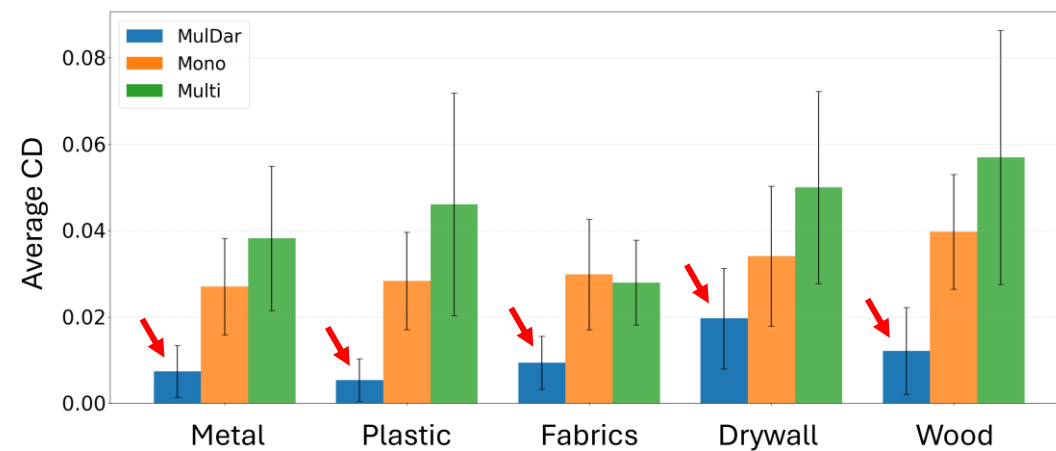
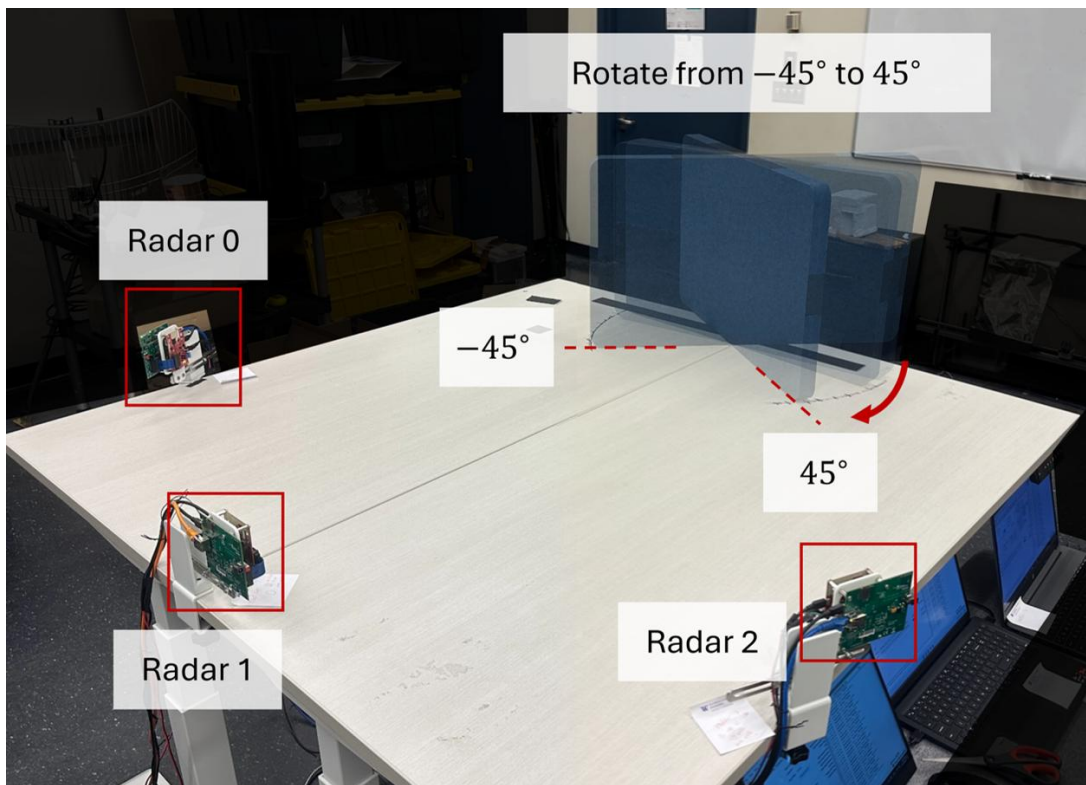


finer reconstruction

(a) Planar scenes

(b) Curved scenes

(c) Deformable & real scenes



MulDar significantly improves the scene reconstruction quality and cuts Chamfer distance to ground truth by **66.79%** over mono-static baselines

What More Can Distributed Radar Achieve



Spatial Channel Selection Ability

What More Can Distributed Radar Achieve



Spatial Channel Selection Ability

Perspective Diversity

What More Can Distributed Radar Achieve



Spatial Channel Selection Ability

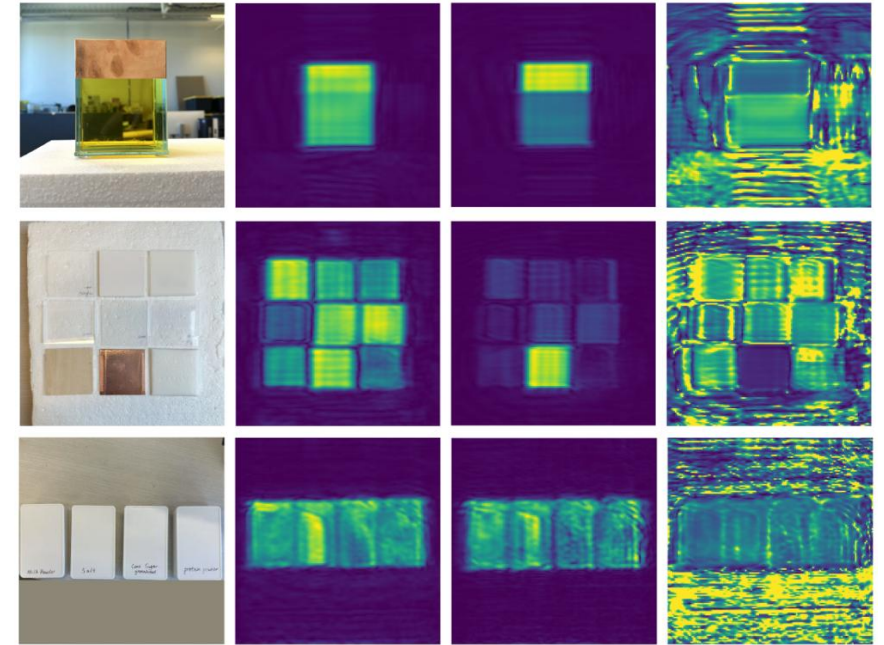
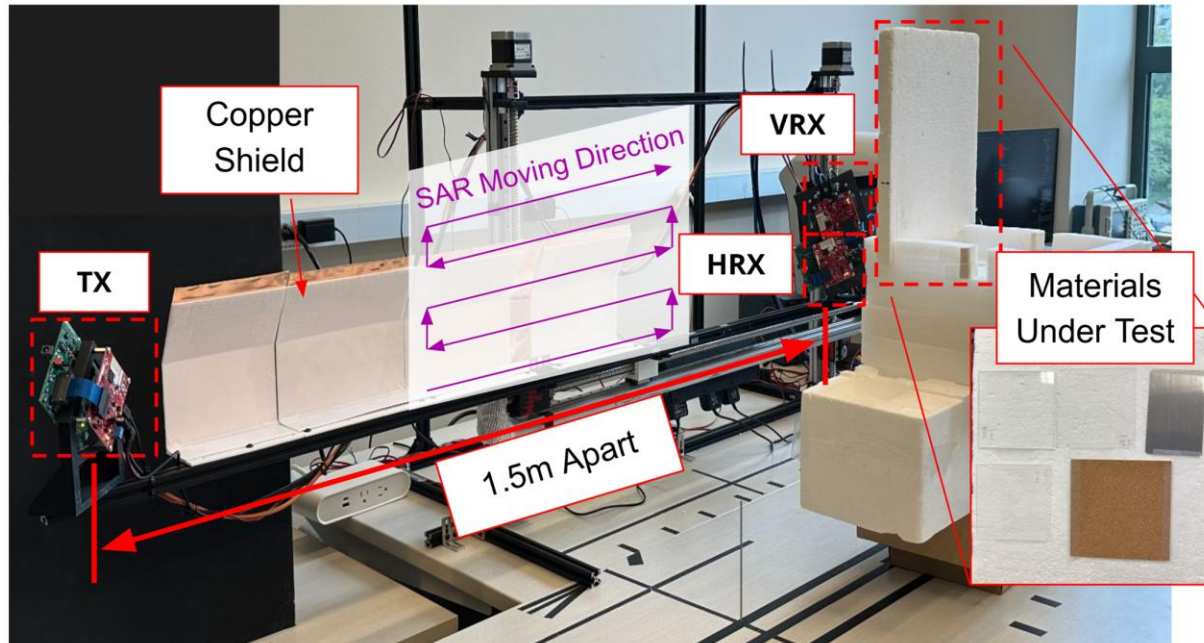
Perspective Diversity



Polarization Diversity

II. Extended Ability with Distributed Radar

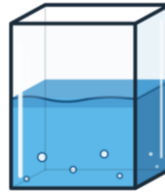
POLySight: Towards Practical Bi-Static Polarimetric Imaging using Commodity mmWave Radars for Material Sensing (SenSys '26)



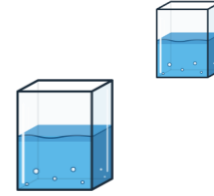
Factors Affecting the Signal



Shape



Size



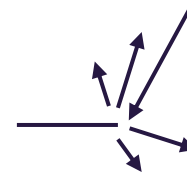
Location



Orientation



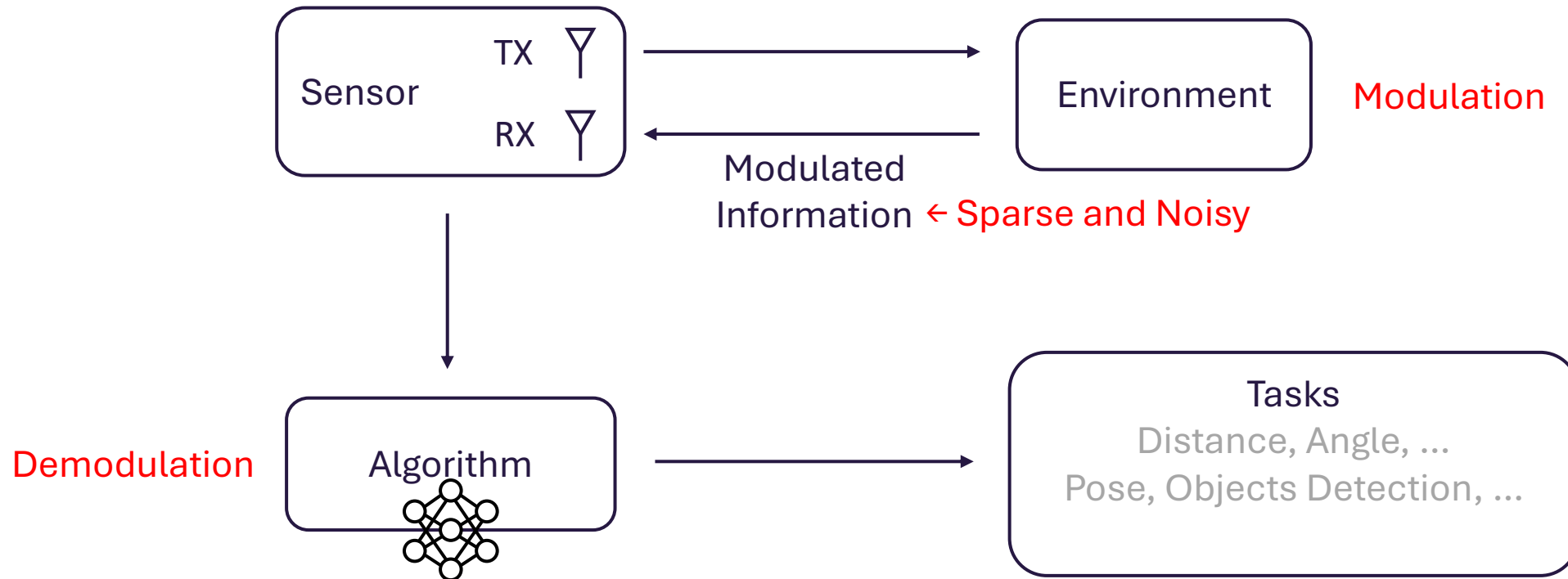
Roughness



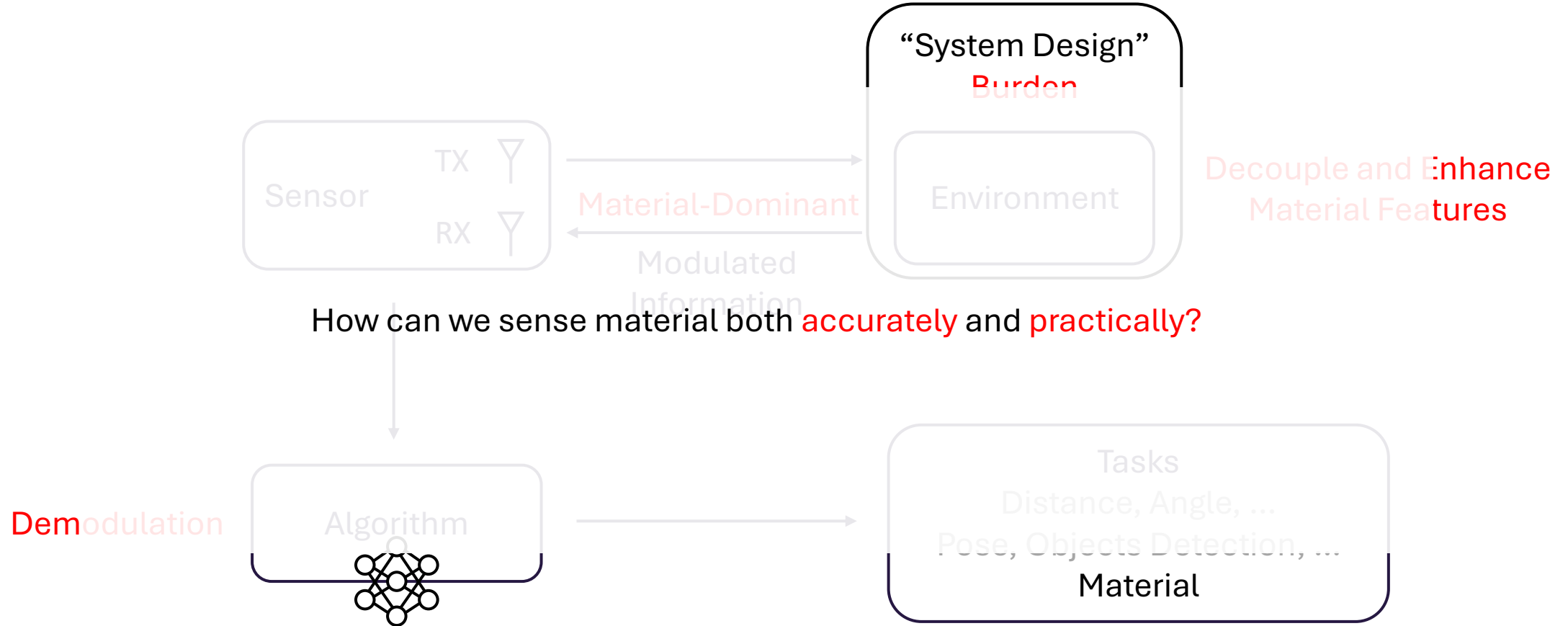
Edge Diffraction

Material isn't the dominant feature

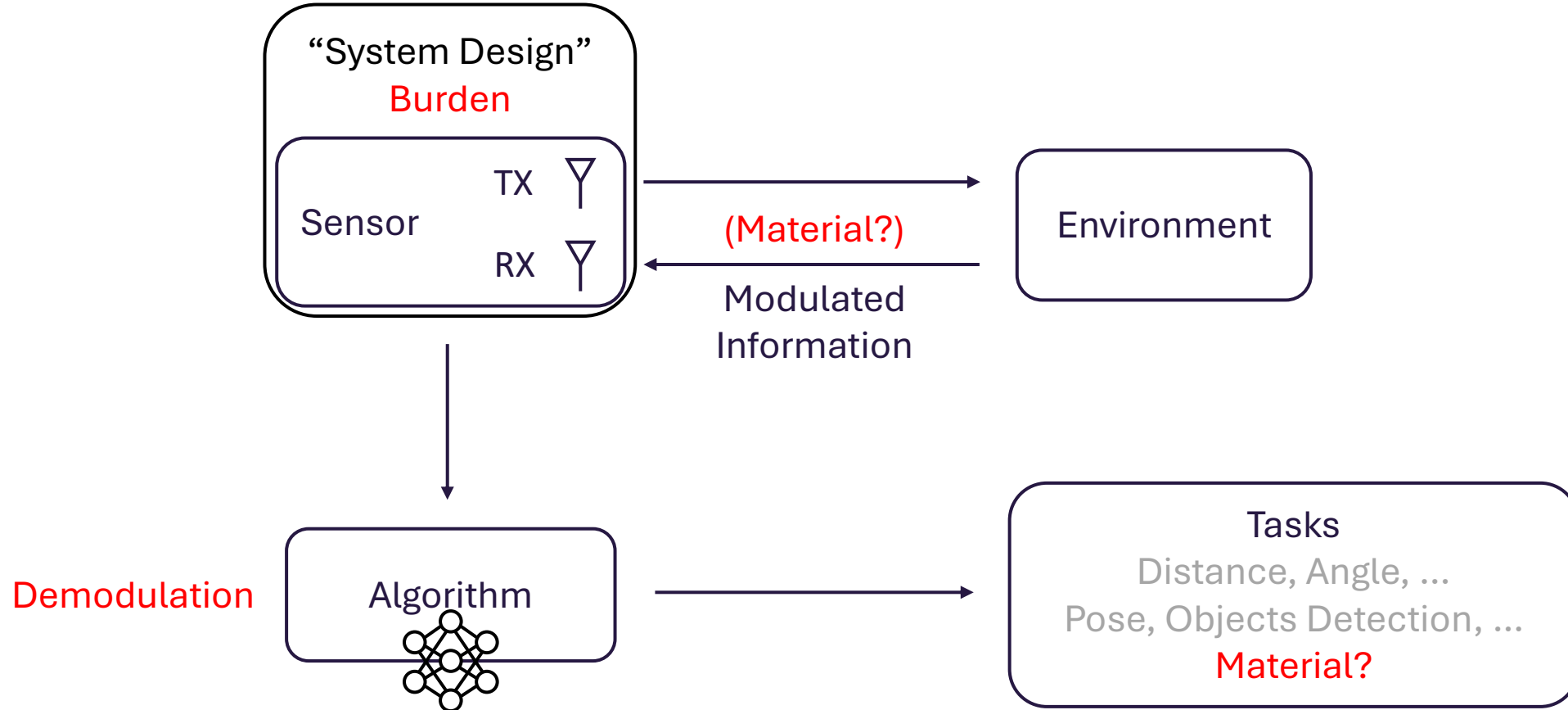
Sensing is Non-cooperative Communication



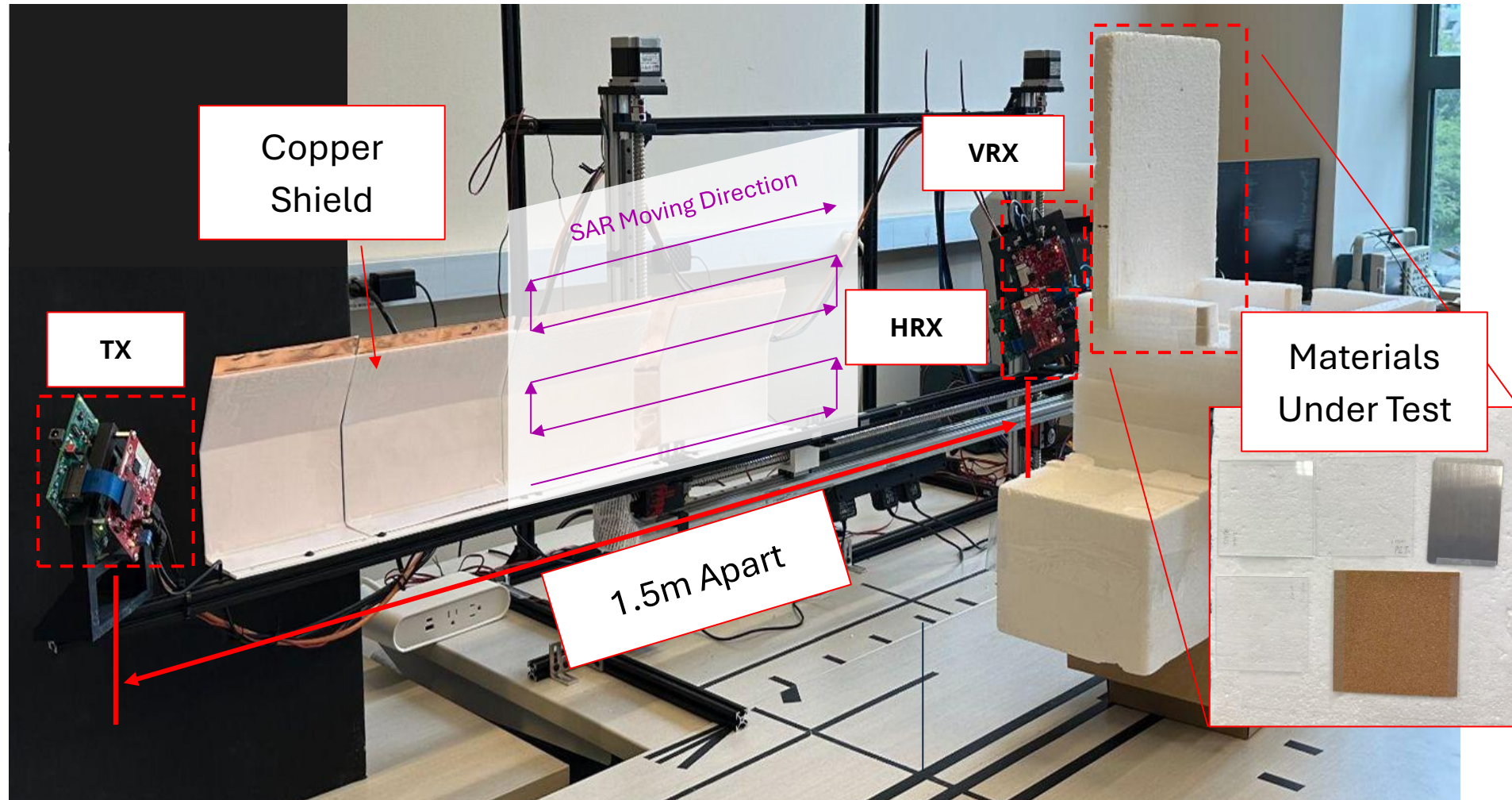
Sensing is Non-cooperative Communication



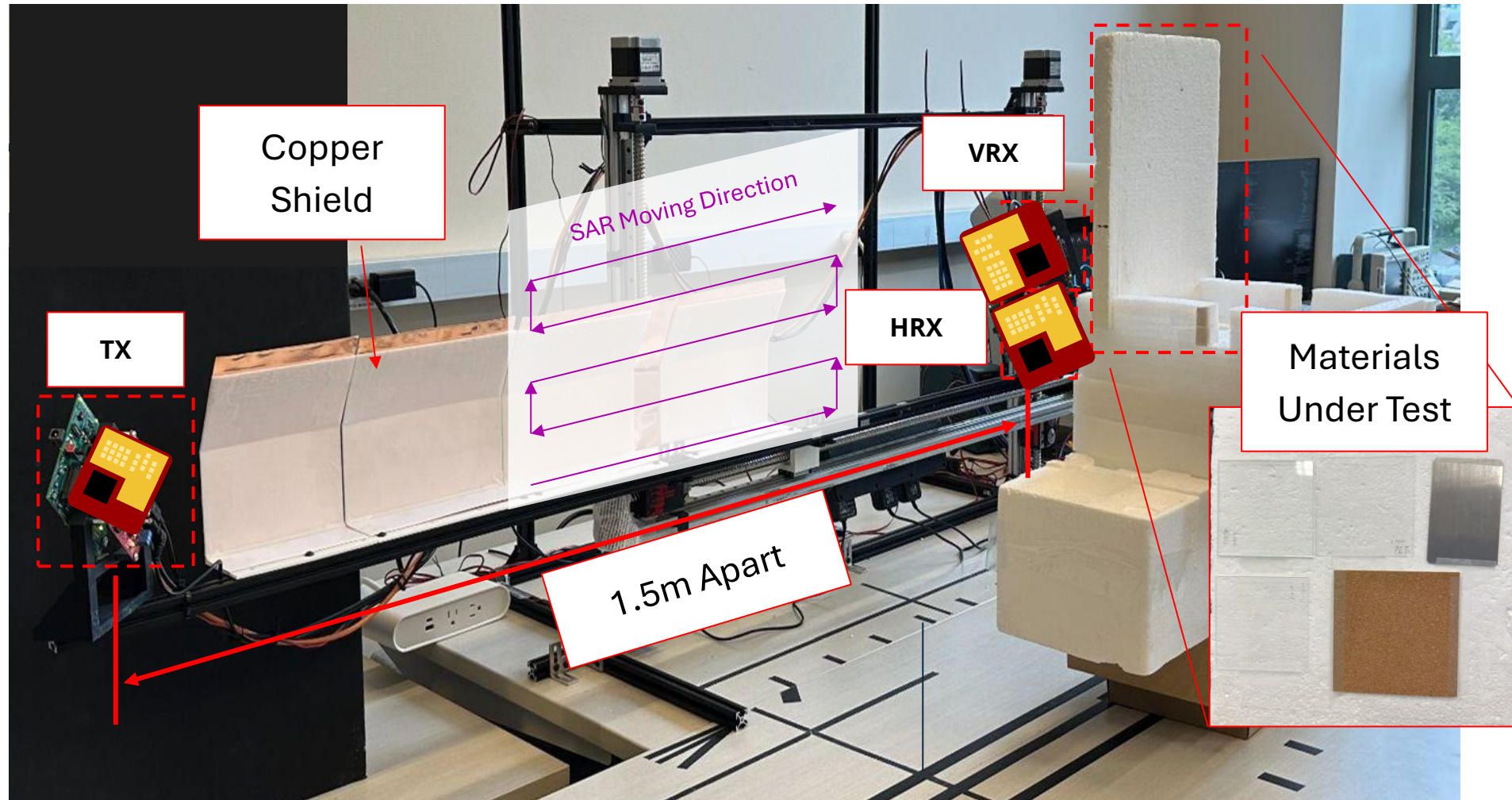
Sensing is Non-cooperative Communication



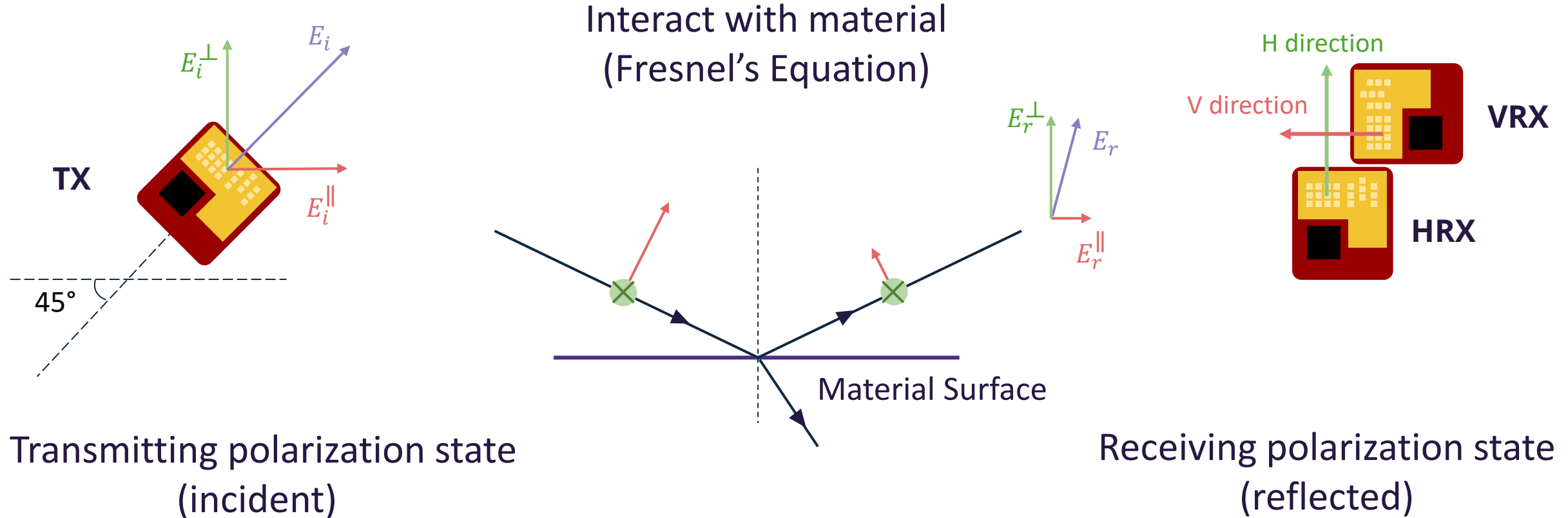
POLySight: Bi-static Polarimetric SAR Imaging System



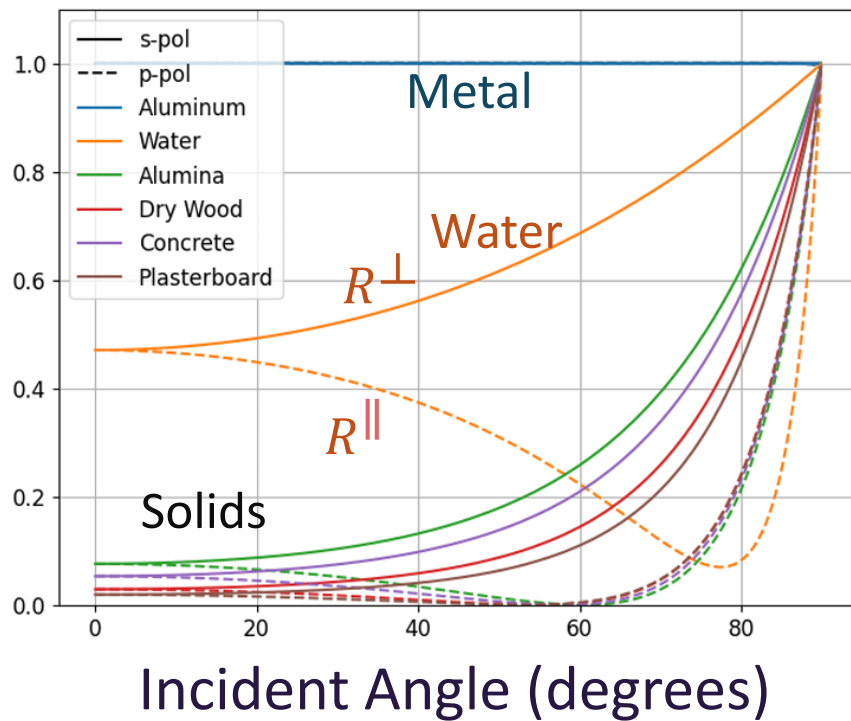
POLySight: Bi-static Polarimetric SAR Imaging System



New Dimension: Polarization

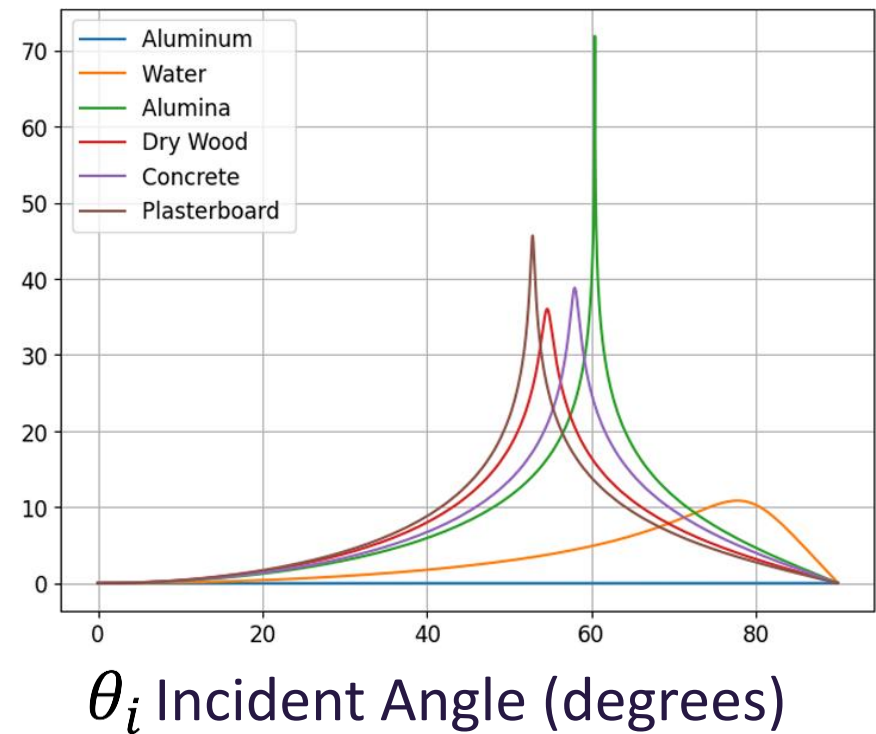


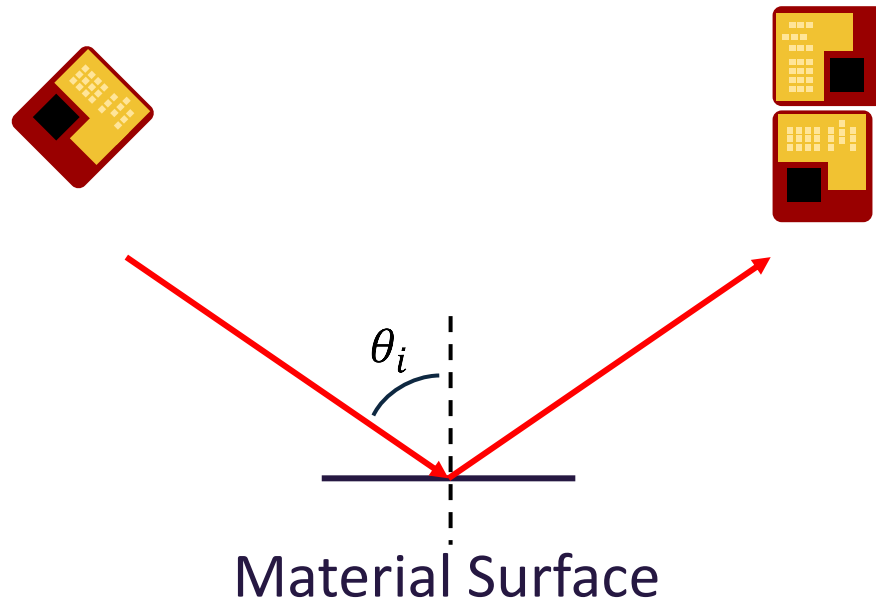
Reflectance $R^\perp$ and $R^\parallel$



Polarization Ratio

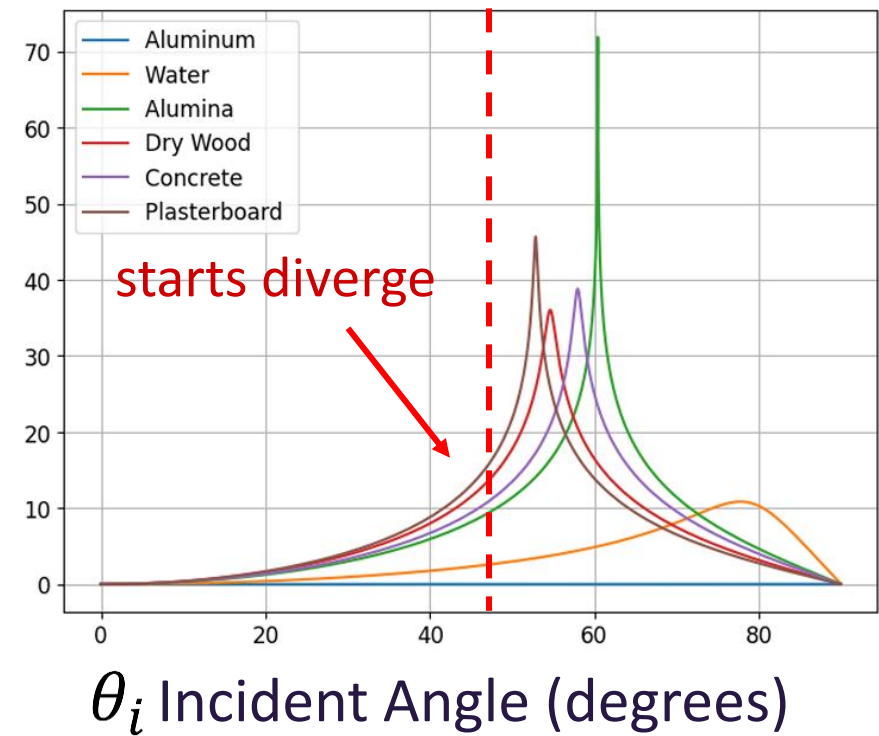
$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$

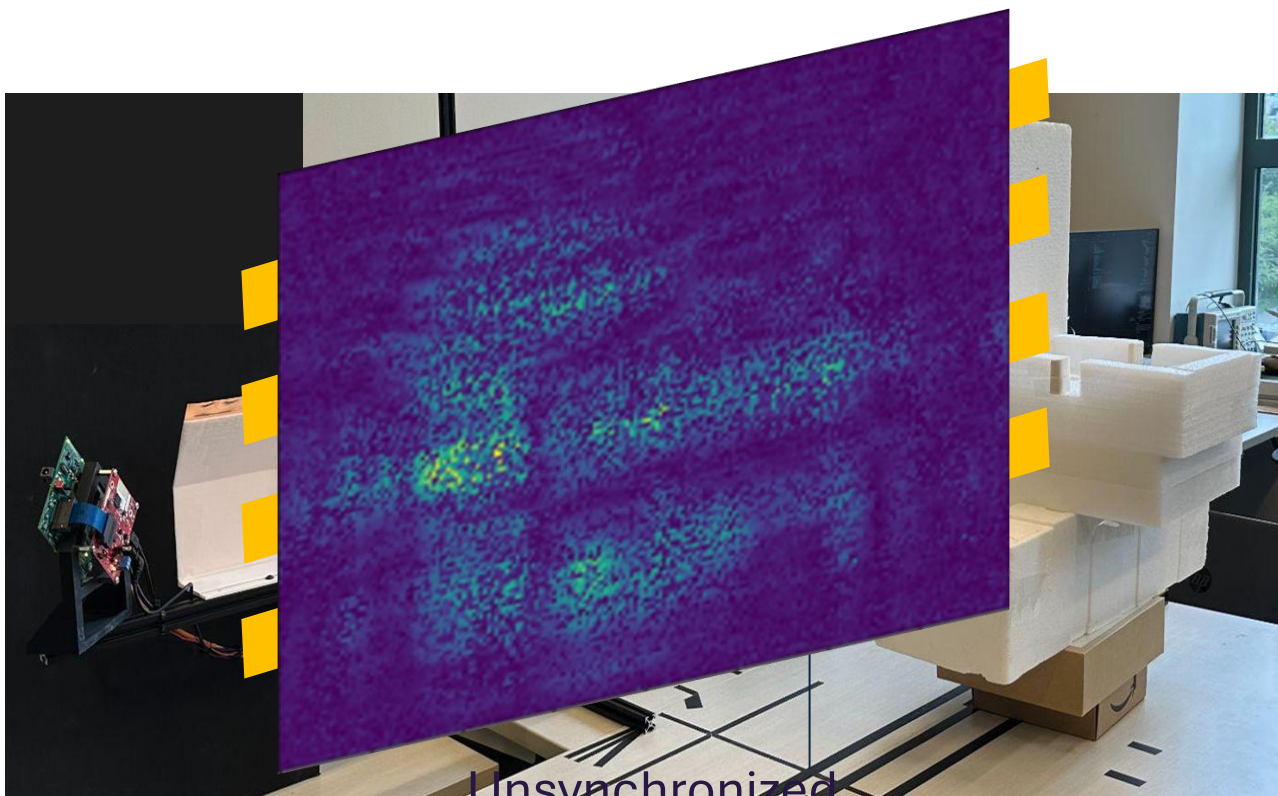




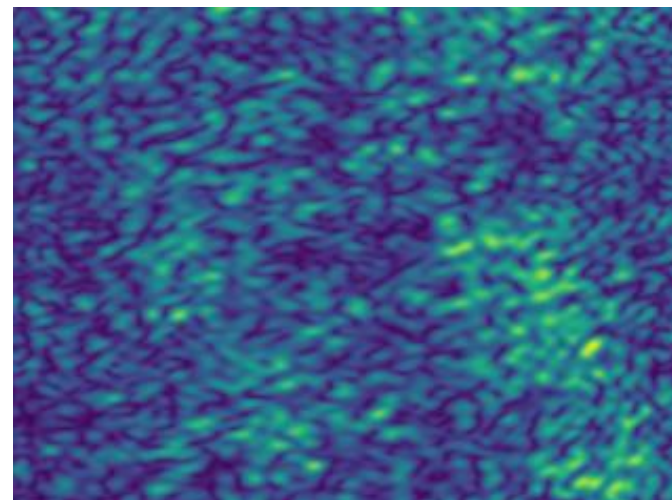
Polarization Ratio

$$|\mathcal{P}| = \left| \frac{R^\perp}{R^\parallel} \right| \text{ (dB)}$$

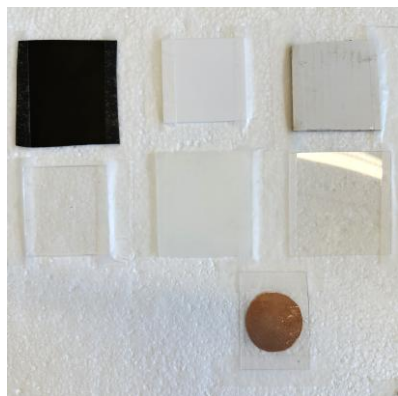




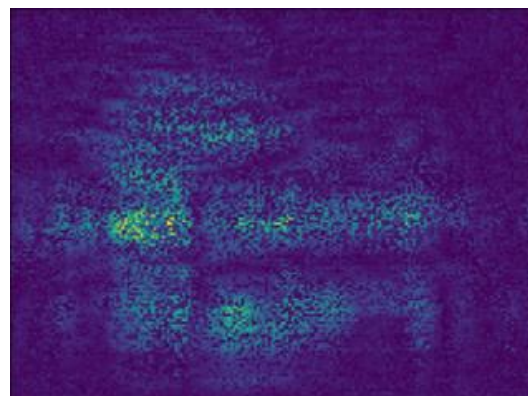
Unsynchronized
Bi-static Radar Cube



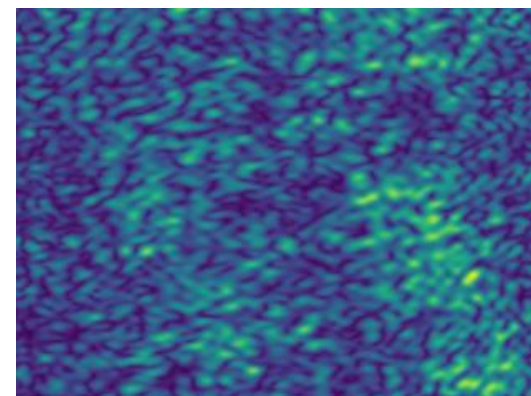
Unsynchronized
Bi-static SAR Image



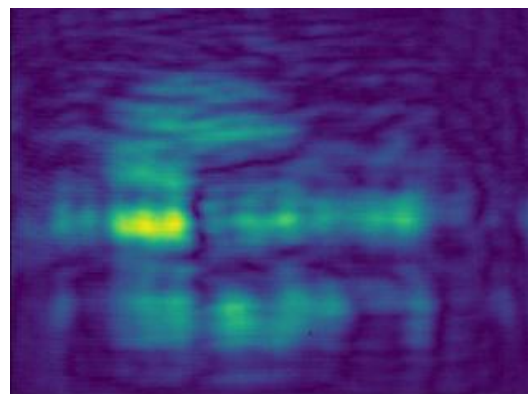
Imaged Materials



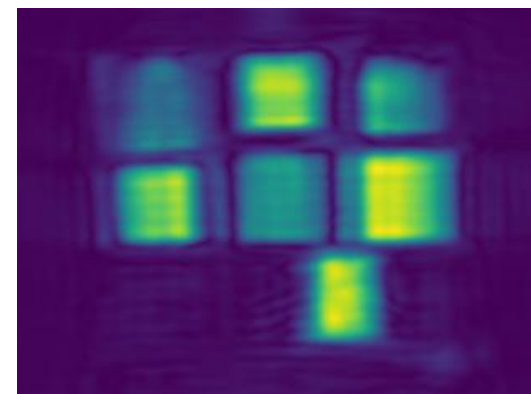
Unsynchronized
Bi-static Radar Cube



Unsynchronized
Bi-static SAR Image



Synchronized
Bi-static Radar Cube



Synchronized
Bi-static SAR Image

Materials

H

V

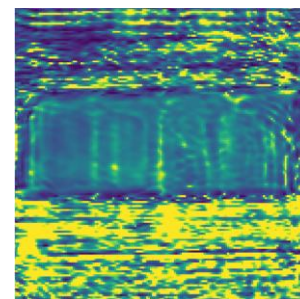
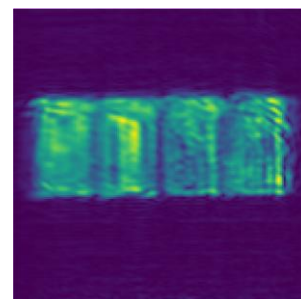
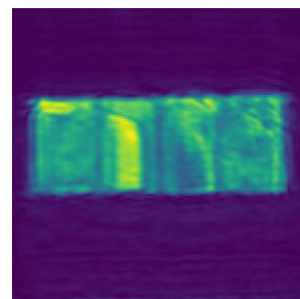
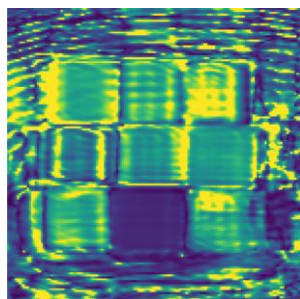
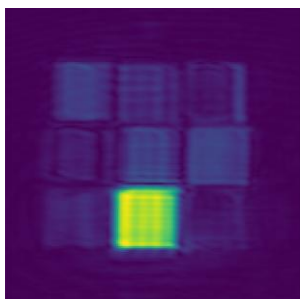
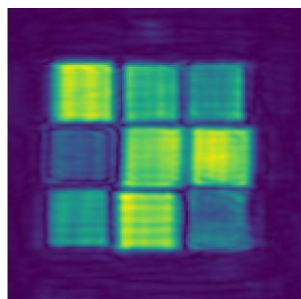
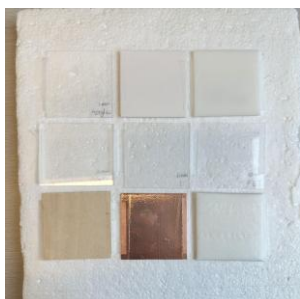
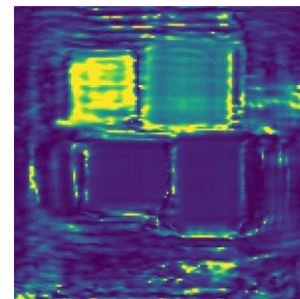
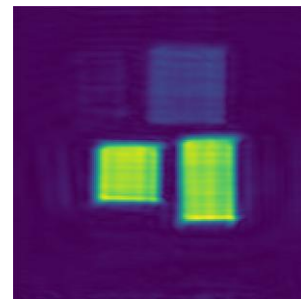
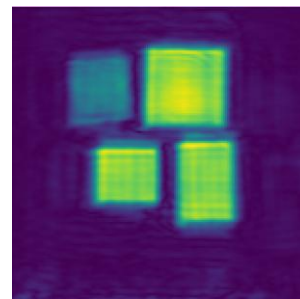
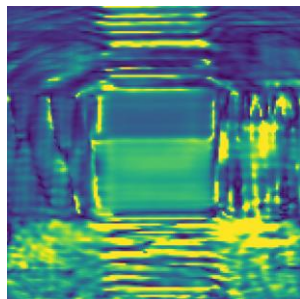
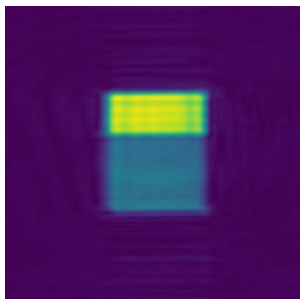
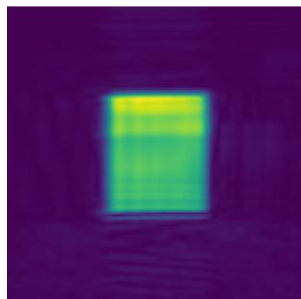
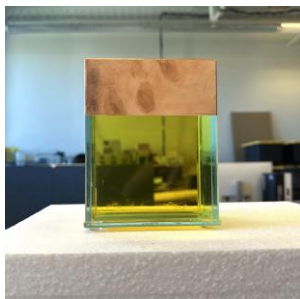
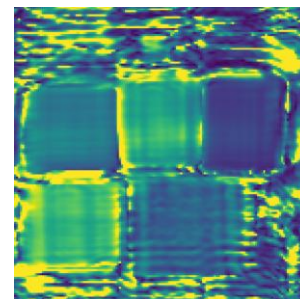
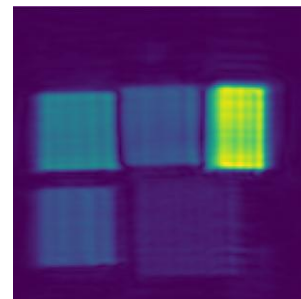
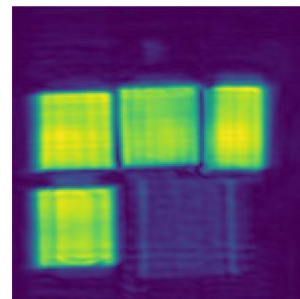
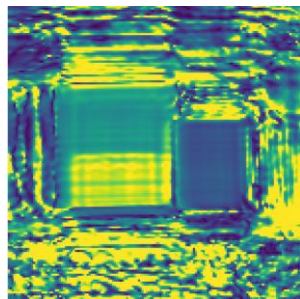
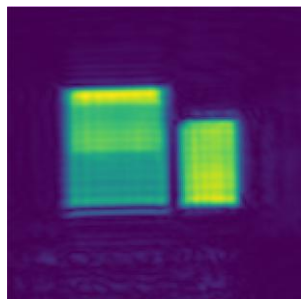
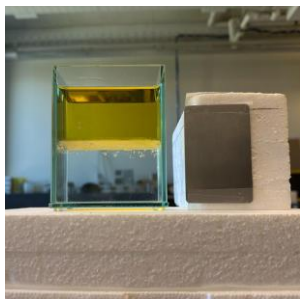
P

Materials

H

V

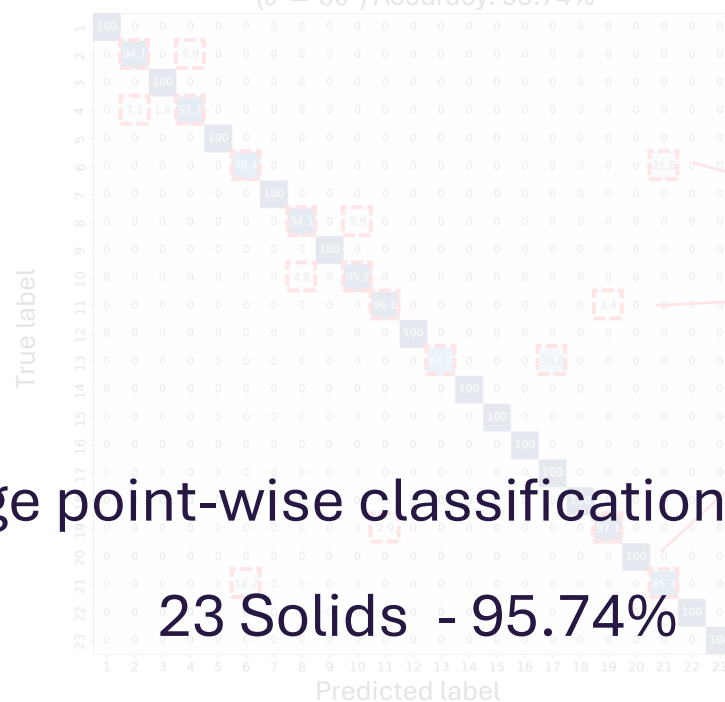
P



w/o/ PolySight

Solids using H channel only
($\theta = 60^\circ$) Accuracy: 48.46%

w/ PolySight

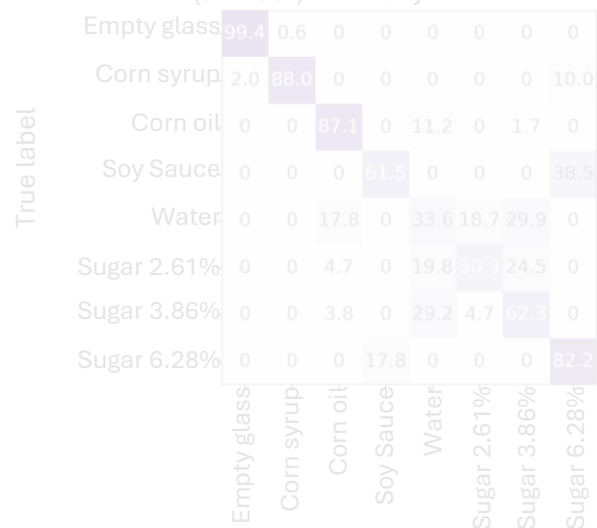
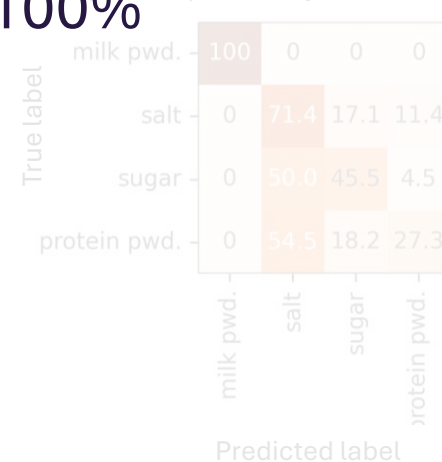
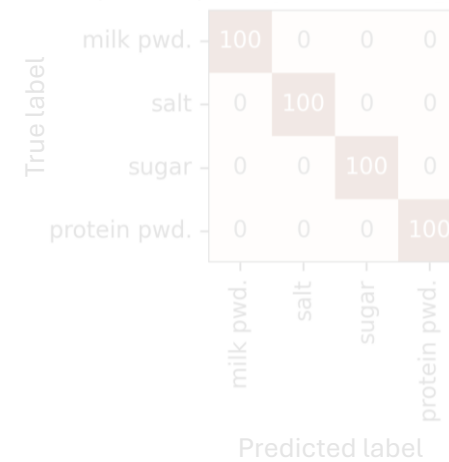
Solids using H, V, P channels
($\theta = 60^\circ$) Accuracy: 95.74%Fine-grained Classification
Solids at multiple incident angles
($\theta = 60^\circ, 55^\circ$) Accuracy: 99.34%

Average point-wise classification accuracy:

23 Solids - 95.74%

8 Liquids - 96.50%

4 Powders - 100%

Liquids using H channel only
($\theta = 55^\circ$) Accuracy: 72.87%Liquids using H, V, P channels
($\theta = 55^\circ$) Accuracy: 96.50%Powders using H channel only
($\theta = 55^\circ$) Accuracy: 58.69%Powders using H, V, P channels
($\theta = 55^\circ$) Accuracy: 100%

w/o/ PolySight

Predicted label

w/ PolySight

Predicted label

w/o/ PolySight

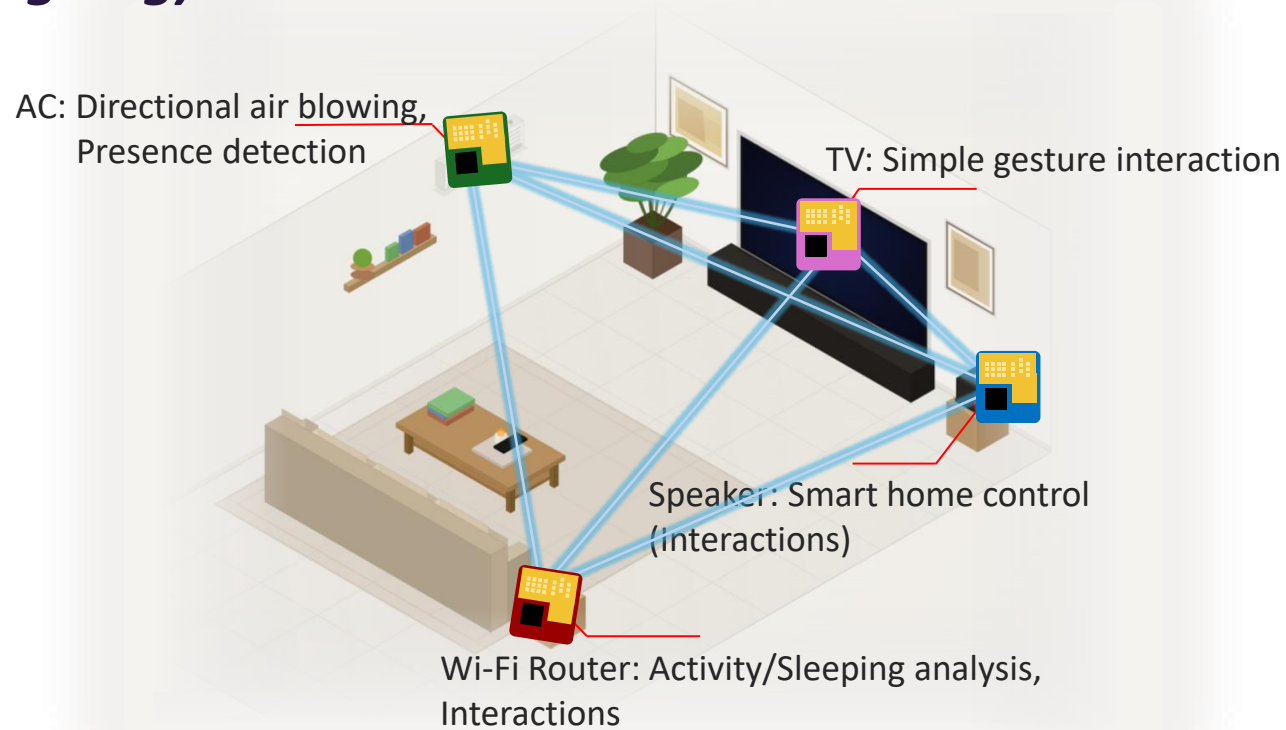
w/ PolySight

Materials	ϵ'	$-\text{j}\epsilon''$
Solids (1–19)		
1. Metal	1.29e5	3.05e5
2. Acrylic (1 mm)	3.56	0.18
3. Acrylic (3 mm)	4.00	0.40
4. Polycarbonate (1 mm)	4.26	0.21
5. PET (0.5 mm)	3.78	0.70
6. PET (1 mm)	3.83	0.73
7. Vinyl (0.37 mm)	3.65	0.70
8. PVC	4.15	0.81
9. PLA (ender, 1 mm)	3.79	0.34
10. PLA (hyper, 1 mm)	4.21	0.30
11. TPU (1 mm)	3.91	0.21
12. Ceramic	4.75	2.43
13. Rubber	5.06	1.11
14. Silicone Rubber	3.52	0.82
15. Glass (2 mm)	7.26	0.85
16. Glass Fiber	4.69	0.70
17. Paper (smooth)	4.63	0.31
18. Cardboard	5.24	0.01
19. Cardboard (rough)	4.54	0.56

Materials	ϵ'	$-\text{j}\epsilon''$
Solids (20–23)		
20. Fabric	2.21	0.13
21. Wood (thin, smooth)	2.61	0.71
22. Wood (rough)	5.02	0.81
23. Cork	10.6	N/A
Liquids		
24. Soy Sauce (w/ glass)	4.30	0.53
25. Corn Oil (w/ glass)	6.01	2.69
26. Corn Syrup (w/ glass)	6.42	0.05
27. Water (w/ glass)	3.66	0.43
28. Cane Sugar (1.3%, w/ glass)	3.72	0.43
29. Cane Sugar (3.8%, w/ glass)	3.78	0.44
30. Cane Sugar (5.1%, w/ glass)	3.81	0.43
31. Cane Sugar (6.3%, w/ glass)	3.89	0.41
32. Cane Sugar (7.4%, w/ glass)	3.96	0.39
Powders		
33. Sugar Powder (w/ plastic)	4.46	0.42
34. Salt (w/ plastic)	4.30	0.94
35. Protein Powder (w/ plastic)	5.74	0.45
36. Milk Powder (w/ plastic)	4.88	0.74

III. Practicality with Distributed Radar

wirelessMulDar (ongoing)



Can they work as a single sensing array?

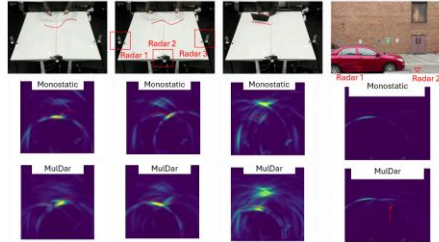
→ Abundant additional information → Practical mmWave Sensing?

Deployment is hard



wirelessMulDar (ongoing) extends MulDar in a wireless and plug-and-play manner for enhanced smart-home mmWave sensing.

Towards Practical mmWave Sensing: A Distributed and Coherent Radar Approach



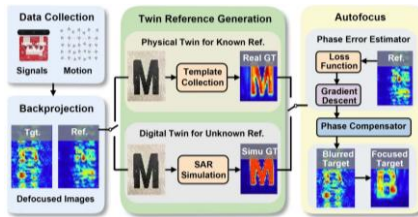
MulDar: Unleashing the Potential of Distributed COTS mmWave Radar by Exploiting Cross-Device Channels

Xinghua Sun, Qiancheng Li, Akshay Gadre

MobiSys 2026, SenSys Demo 2026



MulDar
Code & Dataset & Tutorials



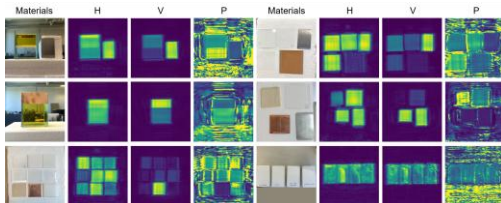
TwinFocus: Autofocus for Handheld mmWave SAR Imaging via Physical and Digital Twin References

Yadong Li, Xinghua Sun, Qiancheng Li, Akshay Gadre

MobiSys 2026, MobiSys Poster 2026



POLySight
Code & Dataset



Poster: Towards Practical Bi-Static Polarimetric Imaging using Commodity mmWave Radars for Material Sensing

Xinghua Sun, Akshay Gadre

MobiSys Poster 2026, SenSys 2026



Questions?
Let's connect!



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